Bellman Calibration for V-Learning in Offline Reinforcement Learning

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Abstract

We introduce *Iterated Bellman Calibration*, a simple, model-agnostic post-hoc procedure for improving off-policy value prediction in infinite-horizon Markov decision processes. Starting from any learned value predictor, the method recalibrates its predictions so that states with similar predicted values have similar average Bellman targets under the target policy. We call this weaker Bellman-style self-consistency property *Bellman calibration*. Unlike full Bellman consistency, it depends only on the predictor's scalar output and can therefore be enforced post hoc. Using off-policy data, we implement this idea by iteratively regressing doubly robust Bellman targets on current predictions via histogram binning or isotonic regression. This yields a simple one-dimensional fitted value iteration that can be applied to any initial value estimator. We establish finite-sample guarantees for calibration and value error under weak assumptions, without requiring Bellman completeness or realizability, and clarify how coverage of the target policy's stationary distribution governs when better calibration improves value prediction.

1. Introduction

Many applications require predicting the long-term consequences of a decision policy in a sequential, stochastic environment. We study this problem in a Markov decision process (MDP), where the goal is to estimate the long-run return of a target policy π, possibly using data collected under a different behavior policy. This off-policy value-prediction problem arises throughout RL and in many application domains: clinicians anticipate long-term health outcomes under proposed treatment rules (Van Calster et al., 2019); online platforms estimate customer lifetime value and retention under alternative recommendation strategies (Theocharous et al., 2015; Maystre & Russo, 2022; Xue et al., 2025); and economists study the downstream effects of counterfactual programs (Rust, 1987; Cowgill & Tucker, 2019). In these settings, decisions often depend on the numerical accuracy of predicted long-run returns. Systematic miscalibration can therefore distort policy comparisons, misstate the size of long-run effects, and lead to poor decisions. Calibration is thus important for personalized forecasting, long-horizon prediction, and reliable policy evaluation.

Modern predictors such as neural networks and gradient-boosted trees are often miscalibrated: their numerical predictions can systematically differ from the quantities they are intended to estimate because of model misspecification, limited data, or distribution shift (Zadrozny & Elkan, 2001; Niculescu-Mizil & Caruana, 2005; Bella et al., 2010; Guo et al., 2017). In reinforcement learning, such errors can be particularly damaging because value estimates are also vulnerable to overestimation, instability, and off-policy bias (Thrun & Schwartz, 2014; Van Hasselt et al., 2016; Fujimoto et al., 2019). This suggests a natural desideratum for value prediction: *calibration*. A value predictor is calibrated if states with similar predicted values have similar expected Bellman targets under the target policy. Equivalently, among states assigned a given predicted value, the average Bellman backup—one-step reward plus discounted continuation value—should match that prediction.

Calibration is therefore a weak but meaningful Bellman-style correctness requirement. It is weaker than full Bellman consistency, since it constrains only the predictor's scalar output rather than the full state-to-value mapping, yet it still rules out systematic under- or overestimation at a given predicted value. This can make value estimates more reliable, interpretable, and less sensitive to harmful extrapolation. While calibration has been widely studied in supervised learning, including classification and regression, where predictions are compared with empirical frequencies or conditional means (Lichtenstein et al., 1977; Platt et al., 1999; Zadrozny & Elkan, 2001; Vovk et al., 2005; Whitehouse et al., 2024), it remains comparatively underexplored for long-horizon value prediction in reinforcement learning.

Long-horizon value prediction brings challenges that are specific to RL and do not arise in standard supervised learning. Most notably, it combines the ingredients of the deadly triad: off-policy learning, function approximation, and bootstrapping (Sutton et al., 1998; Van Hasselt et al., 2018). First, the task is inherently off-policy: we want the value of a target policy π, often using data collected under a different behavior policy. Second, the target itself is not directly

observed. Long-run returns must instead be inferred from one-step transitions, typically through a model or the Bellman equation. Third, because value estimates are defined recursively, approximation errors can propagate through the dynamics and compound over time (Gordon, 1995; Munos & Szepesvári, 2008; Farahmand et al., 2010; Agarwal et al., 2019). Moreover, standard value-learning methods such as temporal-difference learning and fitted value or Q-iteration are iterative approximation procedures, so finite-iteration error, function approximation error, and optimization instability can all affect the final estimate (Tsitsiklis & Van Roy, 1996; Fujimoto et al., 2019; Agarwal et al., 2021). As a result, even widely used RL estimators can yield biased or unstable long-horizon value predictions, and calibration methods from supervised learning cannot be transferred naively to this setting.

These empirical difficulties are mirrored by well-known theoretical challenges in reinforcement learning (Chen & Jiang, 2019; Xie et al., 2022; Meyn, 2024). Analyses of fitted value iteration and related regression-based methods typically rely on strong structural assumptions, such as Bellman completeness, realizability, linear function approximation, or finite state spaces (Gordon, 1995; Baird et al., 1995; Tsitsiklis & Van Roy, 1996; Munos, 2003; Van Roy, 2006; Munos & Szepesvári, 2008; Melo et al., 2008; Bertsekas, 2011; Scherrer, 2014; Fan et al., 2020; Xie & Jiang, 2021; Hu et al., 2025). Bellman completeness requires that the Bellman backup of every function in the class remains in the class, so that each regression step is correctly specified. When this condition fails, the fitted regression problem becomes misspecified, and the resulting projection error can accumulate across iterations, leading to severe degradation in value prediction (Chen & Jiang, 2019; Wang et al., 2021; Foster et al., 2021; Amortila et al., 2020). Although min–max approaches can relax Bellman completeness, they typically introduce additional algorithmic and statistical challenges, including saddle-point optimization, rich critic or dual function classes, and strong coverage requirements (Dai et al., 2018; Uehara et al., 2020; Xie & Jiang, 2020; Xie et al., 2021; Uehara et al., 2021; Zhan et al., 2022; Uehara et al., 2023; Jin et al., 2021; Xie & Jiang, 2021).

Contributions. We study post hoc correction of long-horizon value predictors for a target policy π and establish calibration guarantees under minimal assumptions.

Our main contributions are:

1. We formalize **Bellman calibration**, a weak correctness criterion for infinite-horizon value prediction that is implied by Bellman consistency and satisfied by the true value function. It provides a dynamic analogue of calibration in supervised learning.

2. We introduce **post hoc calibration methods** based on histogram binning and isotonic regression. These can be applied to any learned value predictor, including those produced by fitted value or Q-iteration, and use a doubly robust Bellman target to handle off-policy data.

3. We establish **finite-sample guarantees** for both calibration error and value error. The bounds separate statistical, nuisance-estimation, and finite-iteration errors, and show that our procedure reduces calibration error without sacrificing the best value accuracy achievable by any function of the original predictor's output. The results do not require Bellman completeness or realizability, and they clarify how policy coverage governs when improved calibration translates into improved value accuracy.

Related work. Our work builds on post-hoc calibration methods from supervised learning (Zadrozny & Elkan, 2001; 2002; Niculescu-Mizil & Caruana, 2005; Bella et al., 2010; Guo et al., 2017); see also (Roth, 2022) for a modern overview. Whereas this literature studies calibration of static predictions, we consider calibration of long-horizon value predictions in sequential decision problems through the Bellman residual. Our work is also related to causal calibration for static treatment-effect prediction (Van Der Laan et al., 2023; Whitehouse et al., 2024), which we extend here to dynamic, long-horizon MDPs. Most closely related is van der Laan et al. (2025a), who proposed an isotonic post-hoc calibration method for Q-functions analogous to our approach for V-functions. Their goal, however, was to debias global policy-value estimators in Double RL (Kallus & Uehara, 2020; 2022), rather than to improve value prediction itself. Moreover, calibration of a Q-function does not in general imply calibration of the induced value function, so the target of calibration, the estimand, and the resulting guarantees are different. More distantly related are works on calibrated uncertainty for learned dynamics models (Malik et al., 2019), conformal uncertainty sets in RL (Zhang et al., 2023; Sun et al., 2023), and distributional RL for return distributions (Bellemare et al., 2017; Dabney et al., 2018). These share the goal of improving reliability under uncertainty, but they address different prediction targets from ours.

Many practical variants of fitted Q-iteration and Q-learning aim to improve robustness or approximation quality in offline settings. These include adversarial or minimax formulations (Uehara et al., 2021; 2020; Jin et al., 2021; Uehara et al., 2023; Xie & Jiang, 2021; Di et al., 2023), boosted methods that iteratively fit Bellman residuals (Tosatto et al., 2017), conservative or pessimistic updates (Kumar et al., 2020; An et al., 2021; Di et al., 2023), operator-regularized variants (Farahmand et al., 2009), structural approaches based on linear or low-rank models (Shah et al., 2020), learned representations (Chang et al., 2022), state discretization or aggregation (Peng & Williams, 1993; Van Roy, 2006;

Xie & Jiang, 2021), and distributionally robust methods (Zhou et al., 2021). Broadly, these methods seek to stabilize Bellman updates by imposing additional structure, regularizing the effective operator, or reformulating the learning objective. Our approach is complementary: rather than modifying the fitted iteration itself, we apply a post hoc calibration step to an initial value predictor.

2. Calibration for Value Functions

2.1. Setup and Notation

We consider an MDP with continuous state space $\mathcal{S}$, discrete action space $\mathcal{A}$, initial state distribution ρ, transition kernel $P(s' \mid s, a)$, reward function $r_0(s, a)$, and discount factor $\gamma \in [0, 1)$. Data are collected under a behavior policy $b_0(a \mid s)$, and each sample consists of a transition (S, A, R, S'), where $(S, A) \sim \rho \times b_0$, $S' \sim P(\cdot \mid S, A)$, and $R = r_0(S, A) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid S, A] = 0$. We observe a calibration dataset of n i.i.d. transitions, $\mathcal{C}_n := \{(S_i, A_i, R_i, S_i')\}_{i=1}^n$.

Let $\pi(a \mid s)$ denote a target policy and $w_\pi := \pi/b_0$ the corresponding importance ratio. Define $(\pi f)(s) := \sum_{a\in\mathcal{A}} \pi(a \mid s) f(s, a)$ and $(Pv)(s, a) := \mathbb{E}[v(S') \mid S = s, A = a]$. The policy-marginalized quantities are $P_\pi := \pi P$ and $r_{0,\pi} := \pi r_0$, with $\mathcal{T}_\pi(f) := r_{0,\pi} + \gamma P_\pi f$ the associated Bellman operator. A measure μ is *stationary* for a Markov operator P if $\mu P = \mu$.

The *value function* under policy π is the expected discounted return starting in state s and following π:

$$v_0(s) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t R_t \,\middle|\, S_0 = s\right].$$

Here $\mathbb{E}_\pi$ denotes expectation over $A \sim \pi(\cdot \mid S)$ and $S' \sim P(\cdot \mid S, A)$. It is the unique bounded fixed point of the Bellman equation (Bellman, 1952; 1966)

$$v_0 = \mathcal{T}_\pi(v_0), \quad \mathcal{T}_\pi(v_0)(s) = r_{0,\pi}(s) + \gamma\,(P_\pi v_0)(s). \quad (1)$$

Let $L^2(\rho)$ denote the space of square-integrable functions of S under ρ, with norm

$$\|f\|_{2,\rho} := \{\mathbb{E}_\rho[f(S)^2]\}^{1/2},$$

and for any measure μ on $\mathcal{S} \times \mathcal{A}$, let $\|f\|_{2,\mu} := \{\int f(s, a)^2\, \mu(ds, da)\}^{1/2}$. Let $\|f\|_{n,S,A}$ and $\|g\|_{n,S'}$ denote the empirical L^2 norms over $\{(S_i, A_i)\}_{i=1}^n$ and $\{S_i'\}_{i=1}^n$, respectively. We write $\lesssim$ for inequalities up to absolute constants, and use $[K] := \{0, 1, \ldots, K\}$.

2.2. Bellman Calibration: A Minimal Requirement

Let $\hat{v}$ denote an estimated value function for the target policy π, fit on data independent of the calibration sample $\mathcal{C}_n$. We do not seek to modify the underlying value-learning procedure or estimate a new value function from scratch. Instead, we take an existing predictor and impose a minimal correctness requirement on its outputs. We formalize this through *Bellman calibration*, a weaker, score-based analogue of Bellman consistency.

A natural requirement for a value predictor is *Bellman consistency*, namely

$$\hat{v}(S) \approx \mathbb{E}_\pi[R + \gamma \hat{v}(S') \mid S].$$

Equivalently, the Bellman residual $R+\gamma\hat{v}(S')-\hat{v}(S)$ should have conditional mean zero given the full state. In high-dimensional or continuous state spaces, however, this is a strong requirement: enforcing or even checking it generally requires learning a fully state-dependent conditional mean, often under strong structural assumptions on the function class, such as Bellman completeness (Chen & Jiang, 2019).

We instead consider a weaker requirement that requires correctness only after grouping states by their predicted value:

$$\hat{v}(S) \approx \mathbb{E}_\pi[R + \gamma \hat{v}(S') \mid \hat{v}(S)].$$

That is, among states assigned the same predicted long-run return, the prediction should be Bellman-consistent on average. This is the dynamic analogue of post hoc calibration in supervised prediction, where predictions are required to be correct on average within level sets of a scalar score (Zadrozny & Elkan, 2002; Niculescu-Mizil & Caruana, 2005; Guo et al., 2017). The key benefit is that it reduces a high-dimensional conditional moment restriction to a one-dimensional one, making post-hoc correction feasible without strong structural assumptions or relearning a fully state-dependent value function.

Bellman calibration. For any candidate value function v, define the Bellman-calibration map

$$\Gamma_0(v)(s) := \mathbb{E}_\pi[R + \gamma v(S') \mid v(S) = v(s)]. \quad (2)$$

We say that v is *Bellman calibrated* if

$$v(S) = \Gamma_0(v)(S) \quad \text{almost surely},$$

or equivalently, $\mathbb{E}_\pi[R + \gamma v(S') - v(S) \mid v(S)] = 0$ almost surely. We quantify deviations from this condition by the *Bellman calibration error*

$$\mathrm{Cal}_{\ell^2}(v) := \|v - \Gamma_0(v)\|_{2,\rho}. \quad (3)$$

Bellman calibration is necessary but not sufficient. The true value function v_0 is Bellman calibrated. Indeed, by the Bellman equation,

$$v_0(S) = \mathbb{E}_\pi[R + \gamma v_0(S') \mid S] \quad \text{almost surely}.$$

Applying the tower property and conditioning on the coarser variable $v_0(S)$ gives

$$v_0(S) = \mathbb{E}_\pi[R + \gamma v_0(S') \mid v_0(S)] = \Gamma_0(v_0)(S) \quad \text{almost surely.}$$

Thus, any Bellman-consistent value function is automatically Bellman calibrated. The converse need not hold. Because Bellman calibration conditions only on the scalar score $\hat{v}(S)$, it does not rule out substantial heterogeneity in the true value among states with the same predicted score. As a result, even an inaccurate predictor may be Bellman calibrated if its errors average out within score levels (Gupta et al., 2020). Bellman calibration should therefore be viewed as a minimal reliability requirement, not as a guarantee of overall value accuracy.

Bellman calibration also admits a reward-model interpretation. Define the reward function implied by a value predictor $\hat{v}$ as $\hat{r}_{\pi,\hat{v}}(s) := \hat{v}(s) - \gamma P_\pi \hat{v}(s)$. This is the one-step reward that would make $\hat{v}$ satisfy the Bellman equation under policy π. Bellman calibration of $\hat{v}$ is equivalent to

$$\mathbb{E}_\pi[R - \hat{r}_{\pi,\hat{v}}(S) \mid \hat{v}(S)] = 0.$$

That is, after grouping states by their predicted long-run return, the reward implied by $\hat{v}$ matches the true reward on average within each group. This anchors value predictions to observed rewards and target-policy transitions, helping limit harmful extrapolation. When $\gamma = 0$, this reduces to ordinary regression calibration (Roth, 2022).

Strong Bellman calibration. The definition above calibrates $\hat{v}$ only with respect to its own Bellman target. A stronger alternative would require calibration with respect to the true value function v_0 itself. Specifically, we say that $\hat{v}$ is *strongly Bellman calibrated* if

$$\hat{v}(S) = \mathbb{E}[v_0(S) \mid \hat{v}(S)] \quad \text{almost surely.}$$

Equivalently, among states with the same predicted value, the prediction agrees on average with the true long-run return under π. This notion is substantially stronger, since it requires learning the conditional mean of the unknown target $v_0(S)$ given the score $\hat{v}(S)$. In general, this is not directly available from one-step Bellman targets alone and is typically no easier than the underlying value-learning problem itself (Kallus & Uehara, 2020; 2022). For this reason, we focus on the weaker but practically attainable notion of Bellman calibration defined above.

2.3. Bellman calibration and value error

We now clarify what Bellman calibration buys us for value prediction. Since it is only a score-based correctness property, it does not by itself guarantee that $\hat{v}$ is close to v_0. The result below decomposes the error of $\hat{v}$ into a *refinement* term, capturing information lost by reducing the state to the score $\hat{v}(S)$, and a *calibration* term, capturing failure of Bellman calibration within that score class.

Define the class of score-based transformations

$$\mathcal{F}_{\hat{v}} := \{\theta \circ \hat{v} : \theta : \mathbb{R} \to \mathbb{R}\},$$

and let $\Pi_{\hat{v}}$ denote the $L^2(\rho)$ projection onto this class:

$$\Pi_{\hat{v}} f := \arg\min_{g \in \mathcal{F}_{\hat{v}}} \|f - g\|_{2,\rho}.$$

Thus, $\Pi_{\hat{v}} f$ is the best predictor of f that depends on the state only through the scalar score $\hat{v}(S)$.

Let $P_{\pi,\hat{v}} := \Pi_{\hat{v}} P_\pi$ denote the corresponding aggregated transition operator, i.e.,

$$P_{\pi,\hat{v}} f(s) = \mathbb{E}_\pi[f(S') \mid \hat{v}(S) = \hat{v}(s)].$$

We then define the aggregated Bellman operator $\mathcal{T}_{\pi,\hat{v}} q := \Pi_{\hat{v}} \mathcal{T}_\pi q$, and let $\hat{v}_0 \in \mathcal{F}_{\hat{v}}$ denote its fixed point:

$$\hat{v}_0 = \mathcal{T}_{\pi,\hat{v}} \hat{v}_0. \tag{4}$$

The function $\hat{v}_0$ is the oracle Bellman-calibrated predictor in the score class $\mathcal{F}_{\hat{v}}$.

A1 There exists a stationary measure $\mu_{\hat{v}}$ for $P_{\pi,\hat{v}}$.

Under A1, $\mathcal{T}_{\pi,\hat{v}}$ is a γ-contraction on $L^2(\mu_{\hat{v}})$ (Lemma 1 in Appendix C). In the on-policy setting, where ρ is stationary for P_π, A1 holds with $\mu_{\hat{v}} := \rho$ by Lemma 2 in Appendix C.

Theorem 1 (Calibration–Refinement Bound)**.** *Under A1,*

$$\|\hat{v} - v_0\|_{2,\mu_{\hat{v}}} \;\le\; \frac{1}{1-\gamma}\|\Pi_{\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}} \;+\; \|\hat{v} - \hat{v}_0\|_{2,\mu_{\hat{v}}}.$$

Theorem 1 separates two sources of value error. The *refinement error* $\|\Pi_{\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}}$ measures the information lost by compressing the state to the scalar score $\hat{v}(S)$, and vanishes only if v_0 is recoverable from that score. The *calibration error* $\|\hat{v} - \hat{v}_0\|_{2,\mu_{\hat{v}}}$ measures how far $\hat{v}$ is from the Bellman-calibrated oracle within the same score class. Thus, Bellman calibration cannot recover information lost by the one-dimensional reduction, but it can remove the additional error due to lack of Bellman self-consistency within that score class. This mirrors classical calibration–refinement decompositions in classification and regression (Murphy, 1973; DeGroot & Fienberg, 1983; Van Der Laan et al., 2023; Whitehouse et al., 2024).

If $\hat{v}$ is Bellman calibrated, then $\hat{v} = \hat{v}_0$ and the calibration term vanishes. The remaining error is then exactly the irreducible refinement error associated with the score $\hat{v}(S)$, up to the factor $(1-\gamma)^{-1}$. Thus, once $\hat{v}$ is calibrated, no further post-hoc transformation of the form $\theta \circ \hat{v}$ can improve the bound without changing the underlying score itself.

On-policy vs off-policy. Under on-policy sampling from a stationary distribution, the behavior distribution ρ is stationary for P_π, and hence also for the aggregated kernel $P_{\pi,\hat{v}}$. Therefore we may take $\mu_{\hat{v}} = \rho$, so the decomposition is measured in the same $L^2(\rho)$ norm:

$$\|\hat{v} - v_0\|_{2,\rho} \;\le\; \frac{1}{1-\gamma} \min_{\theta:\mathbb{R}\to\mathbb{R}} \|\theta \circ \hat{v} - v_0\|_{2,\rho} + \|\hat{v} - \hat{v}_0\|_{2,\rho}.$$

In this case, Bellman calibration affects only the second term: it makes $\hat{v}$ as accurate as possible among predictors that depend on the state only through the score $\hat{v}(S)$.

Under off-policy sampling, this simplification breaks down because $\Pi_{\hat{v}}$ is defined in the behavior norm $L^2(\rho)$, whereas Bellman contraction is governed by the stationary norm $L^2(\mu_{\hat{v}})$ of the aggregated kernel. This norm mismatch is the main difficulty in the off-policy setting (Van Roy, 2006; Patterson et al., 2022; van der Laan & Kallus, 2025), and is formalized later in Theorem 4.

2.4. A Doubly Robust Bellman Target

Off-policy prediction requires estimating $\mathcal{T}_\pi(v)$ from data generated under the behavior policy b_0. Importance weighting with $w_\pi = \pi/b_0$ corrects this distribution shift, but can be unstable when the weights are poorly estimated. To obtain a more stable target, we use a doubly robust Bellman pseudo-outcome that combines importance weighting with estimates of the reward and transition functions. This target is unbiased if either the weights or the reward–transition model is correctly specified.

Let $\widehat{w}_\pi$ estimate $w_\pi := \pi/b_0$, let $\hat{r}$ estimate the reward function r_0, and let $\widehat{P}$ estimate the Markov operator P. For any function v, define the doubly robust fitted Bellman target $\widehat{\mathcal{T}}_\pi(v)(S,A,R,S')$ as

$$(\pi\hat{q}_v)(S) + \widehat{w}_\pi(A \mid S)\big[R + \gamma v(S') - \hat{q}_v(S,A)\big],$$
$$\text{where } q_v := r_0 + \gamma P v \text{ and } \hat{q}_v := \hat{r} + \gamma \widehat{P} v.$$

When $\gamma = 0$, this reduces to the doubly robust pseudo-outcome from static treatment-effect estimation (Kennedy, 2023; Van Der Laan et al., 2023).

Recall that $(b_0 f)(s) := \sum_{a\in\mathcal{A}} b_0(a \mid s)\, f(s,a)$ and define the denoised Bellman operator

$$\widehat{\mathcal{T}}_{0,\pi}(v)(s) \;:=\; \mathbb{E}\Big[\widehat{\mathcal{T}}_\pi(v)(S,A,R,S') \,\Big|\, S = s\Big].$$

Theorem 2 (Doubly robust errors)**.** *For any v,*

$$\widehat{\mathcal{T}}_{0,\pi}(v) - \mathcal{T}_\pi(v) = b_0\{(w_\pi - \widehat{w}_\pi)(\hat{q}_v - q_v)\}.$$

The fitted Bellman target is *doubly robust*: it is unbiased whenever $\widehat{w}_\pi = w_\pi$ or $\hat{q}_v = q_v$ (for example, if $\widehat{r} = r_0$ and $\widehat{P} = P$). The ratio w_π is known whenever the behavior policy b_0 is known, and in particular when $b_0 = \pi$, in which case $w_\pi \equiv 1$. Then the pseudo-outcome remains valid even with trivial nuisance estimates $\widehat{r} = 0$ and $\widehat{P}v = 0$, reducing to the standard importance-weighted target

$$\widehat{\mathcal{T}}_\pi(v)(S,A,R,S') = w_\pi(A \mid S)\big(R + \gamma v(S')\big). \quad (5)$$

Conversely, in settings such as robotics or simulation where the transition kernel is known, so that $\widehat{P} = P$, the bias reduces to $b_0\{(\widehat{w}_\pi - w_\pi)(\widehat{r} - r_0)\}$. Thus doubly robustness then applies to the weight and reward estimators alone.

Algorithm 1 Iterated Bellman Calibration

input Value predictor $\hat{v}$, calibration data $\mathcal{C}_n$, fitted Bellman operator $\widehat{\mathcal{T}}_\pi$, calibrator class $\mathcal{F}$, iterations K
1: Initialize $\hat{v}^{(0)} \leftarrow \hat{v}$
2: **for** $k = 0, \dots, K-1$ **do**
    `# Construct Bellman target`
3: $\hat{\chi}_i^{(k)} \leftarrow \widehat{\mathcal{T}}_\pi(\hat{v}^{(k)})(S_i, A_i, R_i, S_i')$
    `# Fit 1D calibrator`
4: $\theta_n^{(k+1)} \leftarrow \arg\min_{\theta\in\mathcal{F}} \sum_{i=1}^n (\hat{\chi}_i^{(k)} - \theta(\hat{v}(S_i)))^2$
    `# Update predictor`
5: $\hat{v}^{(k+1)} \leftarrow \theta_n^{(k+1)} \circ \hat{v}$
6: **end for**
output $\hat{v}^{(K)}$

3. Algorithms and Theory

3.1. Iterated Bellman Calibration Algorithm

Algorithm 1 introduces a simple post-hoc calibration method, ***Iterated Bellman Calibration***, which transforms an initial value predictor $\hat{v}$ into a calibrated predictor of the form $\hat{v}^{\text{cal}} := \theta_n \circ \hat{v}$, where the calibrator $\theta_n : \mathbb{R} \to \mathbb{R}$ is learned from the calibration data $\mathcal{C}_n$. At a high level, the method performs fitted Bellman iteration on the one-dimensional score $\hat{v}(S)$: at each step, it regresses doubly robust Bellman targets $\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(k)})(S_i, A_i, R_i, S_i')\}_{i=1}^n$ onto the initial scores $\{\hat{v}(S_i)\}_{i=1}^n$ using a regression class $\mathcal{F}$, and sets $\hat{v}^{(k+1)} = \theta_n^{(k+1)} \circ \hat{v}$. Relative to standard fitted value iteration (Munos, 2005; Munos & Szepesvári, 2008), the key restriction is that every iterate remains in the score class $\{\theta \circ \hat{v} : \theta \in \mathcal{F}\}$, thereby avoiding high-dimensional function approximation. Since convergence is geometric, only a small number of iterations is needed in practice.

Heuristically, $\hat{v}^{\text{cal}}$ is calibrated because the iteration targets the fixed point $\hat{v}_0$ of the aggregated Bellman operator in (4), which is Bellman calibrated. In finite samples, recovering this fixed point exactly would typically require a realizability condition—namely, that $\hat{v}_0$ belongs to the score class $\{\theta \circ \hat{v} : \theta \in \mathcal{F}\}$—together with Bellman completeness for the fitted regression step. The methods introduced next avoid these strong requirements.

We study assumption-light instantiations of Algorithm 1 based on histogram regression (Stone, 1977) and isotonic regression (Barlow & Brunk, 1972). These extend standard calibration methods—namely, histogram binning (Zadrozny & Elkan, 2001; 2002; Gupta & Ramdas, 2021) and isotonic calibration (Zadrozny & Elkan, 2002; Niculescu-Mizil & Caruana, 2005; Van Der Laan et al., 2023; van der Laan & Alaa, 2025)—from static prediction problems to the dynamic setting, and admit finite-sample Bellman calibration guarantees.

3.2. Histogram Calibration

Histogram calibration learns a piecewise-constant post-hoc transform by grouping states with similar initial predictions. Specifically, we partition the scores $\{\hat{v}(S_i)\}_{i=1}^n$ into B ordered bins $\{I_1, \dots, I_B\}$, formed either by empirical quantiles (*equal-mass* binning) or by uniform discretization of the prediction range (*equal-width* binning). We then apply Algorithm 1 with calibrator class $\mathcal{F}_B := \mathrm{span}\{\mathbf{1}_{I_b} : b \in [B]\}$, so that each iterate is constant within bins. The bins may be data-adaptive, provided the number of bins B is deterministic, possibly growing with n.

Operationally, histogram calibration updates each bin value by averaging the fitted Bellman targets within that bin. At iteration K, the calibrator is piecewise constant:

$$\theta_n^{(K)}(t) = n_b^{-1} \sum_{i:\, \hat{v}(S_i) \in I_b} \hat{\chi}_i^{(K)} \quad \text{for } t \in I_b,$$

where $n_b := \sum_{i=1}^n \mathbf{1}\{\hat{v}(S_i) \in I_b\}$. Thus, each update simply replaces the value on a bin by the average Bellman target over that bin.

This binwise update yields the empirical relation

$$\hat{v}^{(K)}(s) = \mathbb{E}_n\left[\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})(S, A, R, S') \,\middle|\, \hat{v}^{(K)}(S) = \hat{v}^{(K)}(s)\right]$$

where $\mathbb{E}_n$ denotes empirical conditional expectation over the calibration sample $\mathcal{C}_n$. Hence, when the iteration stabilizes ($\hat{v}^{(K)} \approx \hat{v}^{(K-1)}$), the output is approximately an empirical fixed point of the aggregated Bellman equation, and therefore approximately Bellman calibrated.

Calibration error. We next derive finite-sample, finite-iteration bounds on the Bellman calibration error (3).

The following technical conditions simplify the analysis and could be relaxed with additional work.

C1 (*Boundedness*) R, r, $\widehat{r}$, w_π, $\widehat{w}_\pi$, and $\hat{v}$ are uniformly bounded by a constant $M \in (0, \infty)$.

C2 (*Sample splitting*) The estimators $\widehat{r}$, $\widehat{w}_\pi$, $\widehat{P}$, and $\hat{v}$ are obtained from data independent of $\mathcal{C}_n$.

C3 ($\widehat{P}$ *is L–Lipschitz*) For all $f, g \in \mathcal{F}_B$ with $\|f\|_\infty, \|g\|_\infty \le M$, $\|\widehat{P}(f \circ \hat{v}) - \widehat{P}(g \circ \hat{v})\|_{n,A,S} \le L\,\|(f - g) \circ \hat{v}\|_{n,S'}$.

Condition C1 can be relaxed using weight clipping together with appropriate tail conditions on the density ratio and outcomes, following variance-control ideas from importance-weighted RL (Hallak et al., 2016; Han & Sung, 2019; Mehrabi & Wager, 2024) and from heavy-tailed regression. Condition C2 can similarly be relaxed by standard cross-fitting arguments (Van Der Laan et al., 2023; Foster & Syrgkanis, 2023). For classification with $\gamma = 0$, Gupta & Ramdas (2021) show that outcome-agnostic histogram binning allows $\hat{v}$ to be fit on the same data used for calibration. Condition C3 simplifies the analysis by controlling the metric entropy of $\{\widehat{P} f \circ \hat{v} : f \in \mathcal{F}_B\}$ through that of $\mathcal{F}_B$. It holds trivially when $\widehat{P} = 0$, as in the importance-weighted target (5), and it holds with $L = 1$ when $\widehat{P}$ is a discrete nonparametric MLE. When $\widehat{P} = P$, the condition holds at the population level and, under mild regularity conditions, also empirically with high probability.

Theorem 3 (Calibration Error for Histogram Binning)**.** *Assume C1-C3. Then, there exists a $C \in (0, \infty)$ such that, for any $K \in \mathbb{N}$, with probability at least $1 - \delta$,*

$$\begin{aligned}\mathrm{Cal}_{\ell^2}(\hat{v}^{(K)}) \;\le\; & C\left(\sqrt{\frac{B}{n}\log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(1/\delta)}{n}}\right) \\ & + \left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\right\|_{2,\rho} \\ & + \left\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\right\|_{2,\rho}.\end{aligned}$$

The first two terms are the statistical error of one-dimensional histogram calibration: they are exactly the rates one would obtain if the calibration target were observed directly, matching classical bounds for histogram binning (Gupta et al., 2020; Gupta & Ramdas, 2021; van der Laan & Alaa, 2025). The third term is the doubly robust nuisance error from Theorem 2. The final term is the finite-iteration error; under mild conditions it decays geometrically in K (Appendix H.4), as in FVI (Munos & Szepesvári, 2008).

Notably, the leading calibration terms do not depend on the discount factor γ except through the finite-iteration term. This reflects the fact that Bellman calibration is a one-step conditional prediction problem: unlike value estimation, its statistical error does not accumulate over future timesteps. Thus calibration is substantially easier to control statistically. By Theorem 1, however, the extent to which calibration translates into value accuracy still deteriorates as γ increases and distributional coverage weakens, as we now formalize.

Estimation error. The next theorem shows how Bellman calibration affects the value-estimation error of the original predictor $\hat{v}$. In particular, post-hoc calibration does

not introduce any error beyond that already incurred by restricting attention to the histogram score class, and may reduce error due to Bellman inconsistency. This is especially relevant under model misspecification, where calibration can improve self-consistency without relearning a high-dimensional value function.

Define the $L^2(\rho)$ projection onto the histogram score class

$$\Pi_{\hat{v},B} q := \arg \min_{\theta \circ \hat{v} : \theta \in \mathcal{F}_B} \|q - \theta \circ \hat{v}\|_{2,\rho}.$$

Let $\hat{v}_{0,B} \in \{\theta \circ \hat{v} : \theta \in \mathcal{F}_B\}$ denote the unique fixed point of the corresponding aggregated Bellman operator

$$\hat{v}_{0,B} = \mathcal{T}_{\pi,\hat{v},B} \hat{v}_{0,B}, \qquad \mathcal{T}_{\pi,\hat{v},B} q := \Pi_{\hat{v},B} \mathcal{T}_\pi q.$$

The associated aggregated transition operator is $P_{\pi,\hat{v},B} := \Pi_{\hat{v},B} P_\pi$, which satisfies

$$P_{\pi,\hat{v},B} f(s) = \mathbb{E}_\pi \left[f(S') \mid \hat{v}(S) \in I_{\hat{b}(s)} \right],$$

where $\hat{b}(s) = b$ whenever $\hat{v}(s) \in I_b$.

C4 (*Stationary-behavior coverage*) There exists a stationary measure $\mu_{\hat{v},B}$ for $P_{\pi,\hat{v},B}$ and $\hat{\kappa}_B \in (0, \infty)$ such that $\|h\|_{2,\mu_{\hat{v},B}} \le \hat{\kappa}_B \|h\|_{2,\rho}$ for all $h : \mathcal{S} \to \mathbb{R}$.

Theorem 4 (Estimation Error for Histogram Binning)**.** *Assume C1-C4. Then, there exists a $C \in (0, \infty)$ such that, for any $K \in \mathbb{N}$, with probability at least $1 - \delta$,*

$$\begin{aligned}
&\|\hat{v}^{(K)} - v_0\|_{2,\mu_{\hat{v},B}} \\
&\quad \le \gamma^K \|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\quad + \frac{\hat{\kappa}_B}{1-\gamma} \min_{\theta \in \mathcal{F}_B} \|\theta \circ \hat{v} - v_0\|_{2,\rho} \\
&\quad + \frac{\hat{\kappa}_B C}{1-\gamma} \left(\sqrt{\frac{B}{n} \log \frac{n}{B}} + \sqrt{\frac{\log(K/\delta)}{n}} \right) \\
&\quad + \frac{\hat{\kappa}_B}{1-\gamma} \max_{j \in [K]} \left\| (\hat{w}_\pi - w_\pi)(\hat{q}_{\hat{v}^{(j)}} - q_{\hat{v}^{(j)}}) \right\|_{2,\rho}.
\end{aligned}$$

Proof technique. Theorem 4 does requires neither realizability nor Bellman completeness for the original value class. The key point is that Bellman completeness holds automatically for the aggregated Bellman operator $\mathcal{T}_{\pi,\hat{v},B}$. Indeed, defining $\mathcal{F}_{B,\hat{v}} := \{\theta \circ \hat{v} : \theta \in \mathcal{F}_B\}$, we have $P_{\pi,\hat{v},B} f \in \mathcal{F}_{B,\hat{v}}$ for every $f \in \mathcal{F}_{B,\hat{v}}$, since $P_{\pi,\hat{v},B}$ averages over bins of $\hat{v}$ and therefore produces a function that is constant on those bins. Hence $\mathcal{T}_{\pi,\hat{v},B} f \in \mathcal{F}_{B,\hat{v}}$ as well.

The proof therefore proceeds in two steps. First, it shows that $\hat{v}^{(K)}$ converges to the fixed point $\hat{v}_{0,B}$ of the aggregated operator. Second, it controls the gap $\hat{v}_{0,B} - v_0$ via a modification of Theorem 1. Related uses of discretization to induce piecewise-constant classes and avoid Bellman completeness assumptions appear, explicitly or implicitly, in Whitt (1978), Peng & Williams (1993), Van Roy (2006), Chen & Jiang (2019), and Xie & Jiang (2021). Those works discretize the high-dimensional state space and target Bellman consistency, whereas we discretize the score $\hat{v}(S)$, yielding a simpler construction tailored to Bellman calibration.

Discussion. When the importance weights w_π are known and K is sufficiently large, Theorem 3 implies that the calibration error vanishes at rate $\sqrt{(B/n)\log(n/B)}$. By contrast, Theorem 4 shows that the estimation error contains two terms: an approximation term

$$\hat{\kappa}_B (1-\gamma)^{-1} \min_{\theta \in \mathcal{F}_B} \|\theta \circ \hat{v} - v_0\|_{2,\rho},$$

reflecting the loss from restricting attention to binned transformations of $\hat{v}$, and the same stochastic term $\sqrt{(B/n)\log(n/B)}$. Thus the choice of B induces the usual bias–variance tradeoff: small B yields a coarse calibrator and potentially large approximation error, while large B increases variance by leaving too few samples per bin.

To preserve the predictive structure of $\hat{v}$, the binning should be fine enough that $\mathcal{F}_B$ can approximate the identity map well. Indeed,

$$\min_{\theta \in \mathcal{F}_B} \|\theta \circ \hat{v} - v_0\|_{2,\rho} \;\le\; \|\hat{v} - v_0\|_{2,\rho} + \min_{\theta \in \mathcal{F}_B} \|\theta \circ \hat{v} - \hat{v}\|_{2,\rho}. \tag{6}$$

Thus, if $\mathcal{F}_B$ approximates the identity map accurately, the best binned transformation of $\hat{v}$ is nearly as accurate as $\hat{v}$ itself.

For equal-mass binning, choosing $B \asymp n^{1/3}$ yields the worst-case rate (Györfi et al., 2002)

$$\|\hat{v}^{(K)} - v_0\|_{2,\mu_{\hat{v},B}} \;\le\; \frac{\hat{\kappa}_B}{1-\gamma} \|\hat{v} - v_0\|_{2,\rho} + O_p((\log n/n)^{1/3}),$$

and under the same scaling the calibration error also vanishes at rate $(\log n/n)^{1/3}$. This suggests that choosing B to optimize predictive accuracy, for example by cross-validation, will typically also yield good calibration.

Role of coverage in off-policy evaluation. The stationary coverage constant $\hat{\kappa}_B$ in C4 quantifies how well the behavior (sampling) norm controls the stationary norm $L^2(\mu_{\hat{v},B})$ of the target policy π. Smaller values of $\hat{\kappa}_B$ therefore yield sharper guarantees. In the most favorable case, when data are sampled on-policy from the stationary distribution of (P, π), we have $\hat{\kappa}_B = 1$. As the data distribution becomes more off-policy and deviates further from stationarity, $\hat{\kappa}_B$ can grow, reflecting weaker stationary overlap and hence looser error bounds. Thus, Bellman calibration yields the strongest value-accuracy guarantees in the on-policy stationary regime, while accurate value prediction becomes more difficult as off-policy mismatch increases.

3.3. Isotonic Calibration

Histogram calibration requires choosing the number of bins B to balance bias and variance. As a tuning-light alternative, we consider *isotonic iterated Bellman calibration*, obtained by taking $\mathcal{F} := \mathcal{F}_{\text{iso}}$ in Algorithm 1, where $\mathcal{F}_{\text{iso}}$ is the class of nondecreasing functions on $\mathbb{R}$. Each update then performs isotonic regression of the fitted Bellman targets onto the initial scores $\hat{v}(S_i)$, yielding a monotone calibrator $\theta_n^{(k)} \in \mathcal{F}_{\text{iso}}$. This step can be computed in near-linear time via the pool-adjacent-violators algorithm (Best & Chakravarti, 1990).

Like histogram calibration, isotonic calibration pools nearby score levels and replaces them by averaged Bellman targets, but it does so using a data-adaptive partition determined by the monotonicity constraint (Barlow & Brunk, 1972). This provides regularization without requiring a choice of bin number. Because the identity map is monotone, isotonic calibration is less likely to distort an already reliable predictor.

C5 (*Finite variation of target*) There exists $V < \infty$ such that, almost surely, the function $t \mapsto \mathbb{E}[\mathcal{T}_\pi(\hat{v}^{(K)})(S) \mid \hat{v}(S) = t, \mathcal{C}_n]$ has total variation at most V.

Condition C5 is also assumed in Van Der Laan et al. (2023) and is needed to control the calibration function $\Gamma_0(\hat{v})$; for histogram binning, this complexity is governed by B.

Theorem 5 (Calibration Error for Isotonic Calibration)**.** *Assume C1-C5 for $\mathcal{F}_{iso}$. Then, there exists a $C \in (0, \infty)$ such that, for any $K \in \mathbb{N}$, with probability at least $1 - \delta$,*

$$\begin{aligned}\mathrm{Cal}_{\ell^2}(\hat{v}^{(K)}) \;\le\; & C\left(n^{-1/3} + \sqrt{\frac{\log(1/\delta)}{n}}\right) \\ & + \left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\right\|_{2,\rho} \\ & + \left\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\right\|_{2,\rho}.\end{aligned}$$

The isotonic case parallels the histogram analysis, but uses the data-adaptive partition induced by the pool-adjacent-violators algorithm. Importantly, we do not assume that the calibration target is monotone; monotonicity is used only as a regularization device to construct an adaptive partition of $\hat{v}$. The resulting $n^{-1/3}$ rate matches the classical rate for isotonic regression (Chatterjee et al., 2013) and is the same rate achieved by histogram binning with $B \asymp n^{1/3}$, but without requiring the number of bins to be chosen in advance.

3.4. Hybrid Isotonic–Histogram Calibration

Extending the finite-sample estimation analysis from Section 3.2 to isotonic iterated Bellman calibration is not straightforward. Although isotonic regression can be viewed as histogram regression on a data-adaptive partition, the argument in Section 3.2 relies on that partition remaining fixed across iterations. By contrast, applying Algorithm 1 with $\mathcal{F} = \mathcal{F}_{\text{iso}}$ produces a new data-dependent partition at each step. As a result, the iterates no longer correspond to a single aggregated Bellman operator, so the histogram analysis does not apply directly.

Algorithm 2 Iso–Hist Iterated Bellman Calibration

input Value predictor $\hat{v}$, calibration data $\mathcal{C}_n$, fitted Bellman operator $\widehat{\mathcal{T}}_\pi$, iterations K

Stage 1: fit outcome-adaptive partition

$\hat{\chi}_i \leftarrow \widehat{\mathcal{T}}_\pi(\hat{v})(S_i, A_i, R_i, S_i')$

$\theta_{n,\text{iso}} \leftarrow \arg\min_{\theta \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n (\hat{\chi}_i - \theta(\hat{v}(S_i)))^2$

Extract bins $\hat{I}_1, \ldots, \hat{I}_{\hat{B}}$ from flat regions of $\theta_{n,\text{iso}}$

Define $\widehat{\mathcal{F}}_{\hat{B}} := \{\theta : \theta \text{ is constant on each } \hat{I}_b\}$

Stage 2: histogram calibration

Apply Algorithm 1 with class $\widehat{\mathcal{F}}_{\hat{B}}$ to obtain $\hat{v}^{(K)}$

output $\hat{v}^{(K)}$

To retain adaptive binning while recovering a fixed operator structure, we introduce a hybrid isotonic–histogram method (Algorithm 2). The idea is to use one isotonic regression step to learn a data-adaptive partition of the range of $\hat{v}$, then hold that partition fixed and run histogram-based Bellman calibration on top of it. Concretely, we first regress the initial Bellman target on $\hat{v}$ using isotonic regression to obtain an adaptive partition, and then apply Algorithm 1 over this fixed partition for all subsequent iterations. This combines the adaptivity of isotonic calibration with the fixed aggregated-operator structure required by the histogram analysis. Related uses of isotonic regression to construct adaptive partitions appear in conformal prediction (Nouretdinov et al., 2018; van der Laan & Alaa, 2024).

Once the partition is fixed, the guarantees of Section 3.2 apply directly, provided the number of bins produced by the initial isotonic step admits a deterministic bound. More generally, the same reasoning extends to other data-adaptive histogram binning methods, such as regression trees.

Theorem 6 (Calibration and Estimation Error for Alg. 2)**.** *Let $\hat{v}^{(K)}$ denote the output of Algorithm 2. Suppose the isotonic regression step produces at most $\hat{B}$ bins, where $\hat{B} \le B_n$ for some deterministic sequence B_n with probability at least $1 - \delta$. Assume C1-C5 with $B = B_n$. Then, with probability at least $1 - 2\delta$, the guarantees in Theorems 3 and 4 hold for $\hat{v}^{(K)}$ with $B := B_n$.*

Existing isotonic regression theory suggests that the effective number of induced bins satisfies $\widehat{B} = O(n^{1/3})$ when the calibration function is continuously differentiable (Deng et al., 2021); this rate can also be enforced by directly con-

p_{tr}	Model	Full Raw	Iso	Histogram	Hybrid Iso
.5	Boost	**.34±.03**	.40±.06	.40±.06	.40±.06
	Lin	.63±.02	.62±.03	**.62±.03**	.62±.03
	Neur	**.32±.05**	.35±.05	.35±.05	.35±.05
.75	Boost	**.34±.03**	.37±.05	.37±.05	.37±.05
	Lin	.63±.02	**.61±.03**	.61±.03	.61±.03
	Neur	.33±.07	**.31±.04**	.31±.04	.31±.04
.9	Boost	**.34±.03**	.37±.04	.37±.03	.37±.04
	Lin	.63±.02	**.61±.04**	.61±.04	.61±.04
	Neur	.33±.08	**.31±.04**	.33±.04	.31±.04

Table 1. Main results for $n = 50{,}000$ across training proportions p_{tr} under sample splitting. Lower is better.

Iter	Full Raw	Iso	Histogram	Hybrid Iso
10	1.314±0.06	**0.494±0.05**	0.494±0.04	0.497±0.05
25	0.492±0.12	**0.386±0.05**	0.391±0.05	0.388±0.05
50	**0.301**±0.04	0.324±0.03	0.335±0.03	0.324±0.03
100	0.325±0.08	**0.312±0.04**	0.325±0.04	0.312±0.04

Table 2. Neural snapshot performance with sample splitting for $n = 50{,}000$ and training proportion $p_{\text{train}} = 0.9$. Lower is better.

straining the number of constant segments (van der Laan & Alaa, 2024). The B-dependent stochastic term in Theorem 4 scales as $O_p(n^{-1/3})$, consistent with Theorem 5.

For the approximation term in Theorem 4 to be no larger than the initial error $\|\hat{v} - v_0\|_{2,\rho}$ up to constants, the induced histogram class must also approximate the identity map on the range of $\hat{v}$. That is, we require $\min_{\theta \in \widehat{\mathcal{F}}_{\widehat{B}}} \|\theta \circ \hat{v} - \hat{v}\| \approx 0$, so that step functions in $\widehat{\mathcal{F}}_{\widehat{B}}$ nearly preserve the original predictions. This condition is plausible when $\hat{v}$ is already close to v_0, in which case isotonic regression should produce a near-identity transformation. By contrast, when $\hat{v}$ is substantially miscalibrated, isotonic regression may induce a coarser partition that pools regions of poor prediction into a common calibrated value.

4. Illustrative Experiments

We use synthetic data to study the proposed calibration methods in a controlled setting. Our goal is to illustrate their behavior rather than to provide a comprehensive benchmark. Broad empirical evaluations of calibration are well developed in supervised learning (Zadrozny & Elkan, 2002; Guo et al., 2017), whereas such an evaluation in reinforcement learning is beyond the scope of this paper.

Synthetic environment. We evaluate all calibration methods with $K = 1000$ in a synthetic subscription MDP designed to mimic monthly retention and revenue dynamics (Appendix B). We simulate offline data from $n_{\text{cust}} = 50{,}000$ customers over $T = 24$ months with $\gamma = 0.99$ under a fixed heuristic behavior policy, and evaluate off-policy value estimation for a deterministic aggressive target policy, with importance weights in $(0.1, 0.99)$. Performance is measured by $(1-\gamma)$-scaled RMSE on initial states, using Monte Carlo rollouts as ground truth and averaging over 50 runs. Additional experimental settings are reported in Appendix B. For quantile binning, we choose B so that each bin contains at least 200 observations.

Results. Table 1 reports off-policy value estimation error under sample splitting, where a proportion p_{tr} of the data is used to fit the base estimator and the remainder for calibration. In all tables, calibrated estimators are compared to a *Full Raw* baseline, the same estimator retrained on the full dataset without calibration. When entries match to two decimals, boldface indicates the lowest unrounded value. Performance depends strongly on the train-calibration split. When $p_{\text{tr}} = .5$, calibration generally does not improve on Full Raw, likely because the base estimator is fit on too little data. In contrast, for $p_{\text{tr}} = .75$ and $p_{\text{tr}} = .9$, isotonic and hybrid isotonic calibration consistently improve the linear and neural estimators. The boosted estimator remains strongest without further calibration, suggesting that this tree- or histogram-based method is already reasonably well calibrated here. Across settings, hybrid and isotonic calibration perform similarly.

Table 2 examines neural snapshots for $p_{\text{tr}} = .9$. Calibration yields large gains early in training, substantially reducing error at iterations 10 and 25, when the raw estimator is still poorly calibrated. At iteration 50, however, the raw snapshot is best, suggesting that splitting can hurt when the base model is already strong. By iteration 100, isotonic and hybrid calibration again achieve the lowest error. Overall, Bellman calibration appears most useful for under-trained or moderately misspecified estimators, with smaller and less uniform gains once the base estimator is already well tuned.

5. Conclusion and Future Work

We introduced *Iterated Bellman Calibration*, a post-hoc method for improving the reliability of off-policy value predictors in infinite-horizon MDPs. Our analysis gives finite-sample guarantees and convergence to a aggregated Bellman fixed point under weak assumptions, without Bellman completeness or realizability. It also highlights the importance of sampling geometry: calibration is most favorable under on-policy sampling from the target policy's stationary distribution, where the Bellman operator and calibration step are naturally aligned, and becomes harder as off-policy distribution shift grows.

Several directions follow naturally. One is to reduce off-policy mismatch by calibrating under distributions closer to the target stationary distribution, for example through stationary-density-ratio weighting or related emphatic-style corrections (van der Laan & Kallus, 2025; Mahmood et al.,

2015; Patterson et al., 2022). Another is to move beyond binning-based calibrators. Although histogram and isotonic methods are simple and theoretically convenient, they do not extrapolate beyond the support of the calibration data. Under stronger distribution shift, smoother second-stage calibrators based on low-dimensional parametric maps, such as temperature-scaling-type corrections (Platt et al., 1999; Jiang et al., 2011; Guo et al., 2017), may be more effective. More broadly, an important next step is a systematic empirical study of calibration in RL across on-policy and off-policy regimes.

Impact Statement.

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Additional related work

To our knowledge, prior work has not defined or enforced calibration of long-horizon value predictions in RL. Sequential calibration in forecasting and online prediction concerns one-step predictive accuracy and does not impose Bellman consistency (Vovk et al., 2005). Bellman conformal inference (Yang et al., 2024) constructs calibrated prediction intervals for time-series forecasting rather than calibrating value functions in MDPs.

B. Experimental results and info

B.1. Synthetic CRM environment

We evaluate all methods in a synthetic customer-relationship-management (CRM) Markov decision process (MDP) that mimics monthly retention and revenue dynamics for a subscription service. The state $s_t \in \mathbb{R}^6$ encodes tenure in months, engagement, fatigue, value segment, price sensitivity, and an indicator of whether the customer is active. At each time t, the agent selects one of three actions: no promotion, light promotion, or strong promotion. Rewards correspond to monthly revenue after discounts and are zero after churn.

Transitions capture key CRM effects. Churn probability depends on tenure, engagement, fatigue, and action via a logistic model; visit probability depends on engagement, fatigue, and action; and revenue conditional on visit scales a baseline value segment by an action-dependent uplift and a price-sensitivity effect, with log-normal noise. Engagement decays over time but is boosted by successful promotions, while fatigue increases with promotion intensity and decays slowly otherwise. Once a customer churns or reaches a maximum tenure of 60 months, the process enters an absorbing state.

We simulate offline datasets of $n_{\text{cust}} = 50{,}000$ customers over a horizon of $T = 24$ months with discount factor $\gamma = 0.99$. Behavior data are generated by a fixed heuristic policy that sends light promotions by default, suppresses promotions for highly engaged and fatigued customers, and occasionally sends strong promotions to low-engagement, high-value customers. We evaluate off-policy value estimation for a deterministic *aggressive* revenue-seeking target policy that sends strong promotions to low-engagement, high-sensitivity customers and avoids promotions for highly engaged, high-value customers.

Ground-truth values $V^\pi(s_0)$ are estimated via Monte Carlo rollouts in the same environment, and we report $(1-\gamma)$-scaled RMSE between the estimated and Monte Carlo values at initial states, averaged across 50 independent runs.

Base estimators. We consider three base estimators: boosted FQE, linear FVI, and neural FVI. The linear model uses a hand-crafted semi-structured feature map consisting of the six raw state variables, normalized and log-transformed tenure, squared terms for engagement, fatigue, value segment, and price sensitivity, three state-only interaction terms, and additive action one-hot indicators, with no state-action interactions. The neural model takes the raw state concatenated with the action as input and uses a feedforward ReLU network with three hidden layers of width 256. Both FVI estimators are trained by iterating one-step Bellman regressions: the linear model uses ridge-regularized least squares, while the neural model is trained for 100 outer iterations using Adam with learning rate 0.005, one epoch per iteration, and mini-batch size 2000. We do not tune the number of neural iterations, since part of the goal is to illustrate a realistic setting in which a base estimator remains poorly calibrated. Selected neural iterates are saved for the snapshot comparisons. Both FVI estimators use an internal 80/20 train-validation split. The boosted baseline is implemented as a LightGBM FQE model trained on the concatenated state-action input. Under sample splitting, base estimators are fit on the training subset and Bellman calibrators on the held-out calibration subset; we also retrain each base estimator on the full dataset to obtain the *Full Raw* baselines.

For the calibration methods, we use a large fixed number of iterations (1000) and verify convergence by visual inspection.

B.2. Calibrating on full data: no sample splitting

Table 3 reports the main results when calibration is applied on the full dataset, without sample splitting. Table 4 complements this comparison by showing neural FVI snapshot performance across iterations in the same no-splitting setting.

Model	Full Raw	Iso	Histogram	Hybrid Iso
Boost	**0.34±0.03**	0.38±0.07	0.39±0.08	0.38±0.07
Lin	0.63±0.02	**0.62±0.02**	0.62±0.02	0.62±0.02
Neur	0.33±0.08	0.30±0.04	0.30±0.04	**0.30±0.04**

Table 3. Main results for $n = 50{,}000$ without sample splitting, where calibration is performed on the full dataset. Lower is better.

Iter	Full Raw	Iso	Histogram	Hybrid Iso
10	1.293±0.07	0.478±0.05	0.480±0.05	**0.478±0.05**
25	0.452±0.09	**0.347±0.05**	0.347±0.05	0.347±0.05
50	0.324±0.06	0.298±0.04	0.298±0.04	**0.298±0.04**
100	0.325±0.08	0.300±0.04	0.300±0.04	**0.300±0.04**

Table 4. Neural snapshot performance without sample splitting for $n = 50{,}000$ and training proportion $p_{\text{train}} = 1.0$. Lower is better.

B.2.1. Neural snapshots at other splits

Tables 5 and 6 report neural FVI snapshot performance under sample splitting for the additional training proportions $p_{\text{train}} = 0.5$ and $p_{\text{train}} = 0.75$. Together, they show how the effect of calibration evolves over training and how it depends on the amount of data reserved for fitting versus calibration.

Iter	Full Raw	Iso	Histogram	Hybrid Iso
10	1.309±0.06	**0.581±0.05**	0.582±0.05	0.581±0.05
25	0.491±0.13	**0.456±0.06**	0.456±0.06	**0.456±0.06**
50	**0.320±0.07**	0.381±0.05	0.382±0.05	0.382±0.05
100	**0.316±0.05**	0.350±0.05	0.351±0.05	0.350±0.05

Table 5. Neural snapshot performance with sample splitting for $n = 50{,}000$ and training proportion $p_{\text{train}} = 0.5$. Lower is better.

Iteration	Full Raw	Iso	Histogram	Hybrid Iso
10	1.28±0.05	0.52±0.05	**0.52±0.05**	0.52±0.05
25	0.44±0.12	**0.38±0.03**	0.38±0.04	0.38±0.04
50	0.34±0.06	**0.33±0.04**	0.33±0.04	0.33±0.04
100	0.33±0.07	**0.31±0.04**	0.31±0.04	0.31±0.04

Table 6. Neural FVI snapshot results for $n = 50{,}000$ with $p_{\text{train}} = 0.75$. Lower is better.

B.2.2. Comparing to the uncalibrated (raw) model without retraining on the full data

Table 7 compares calibrated estimators to the corresponding raw model fit only on the training split, so any improvements reflect calibration rather than retraining on the full dataset. This comparison is relevant when retraining on all available data is costly or when a gold-standard calibration dataset is available, such as recent trajectories or on-policy data, alongside a larger historical dataset used for initial fitting. Table 8 further examines this comparison for the neural method across training iterations, illustrating how the effect of calibration evolves over the course of fitting.

n	Model	Raw	Iso	Hybrid Iso
10,000	Boost	0.681±0.1	**0.671±0.1**	0.697±0.2
	Lin	0.640±0.04	**0.612±0.05**	0.641±0.05
	Neur	0.582±0.2	0.550±0.2	**0.520±0.1**
50,000	Boost	0.433±0.06	**0.429±0.06**	0.438±0.08
	Lin	0.625±0.02	**0.614±0.03**	0.616±0.03
	Neur	0.419±0.1	0.383±0.08	**0.374±0.08**
100,000	Boost	0.360±0.05	0.358±0.05	**0.353±0.07**
	Lin	0.616±0.02	**0.605±0.02**	0.606±0.02
	Neur	0.379±0.09	0.351±0.05	**0.342±0.05**

Table 7. Main results with sample splitting at training proportion $p_{\text{train}} = 0.5$. The raw estimator is fit only on the training split to isolate the effect of calibration.

Iter	Raw	Iso	Quantile	Hybrid Iso
10	1.424±0.1	1.330±0.1	**0.735±0.2**	0.737±0.2
25	0.665±0.1	0.646±0.1	0.577±0.1	**0.571±0.1**
50	0.612±0.2	0.585±0.2	0.555±0.1	**0.547±0.1**
100	0.582±0.2	0.550±0.2	0.524±0.1	**0.520±0.1**

Table 8. Neural FVI snapshot performance with sample splitting at training proportion $p_{\text{train}} = 0.5$. The raw estimator is fit using only the training split, so differences reflect the effect of calibration rather than access to additional data.

C. Stationary Measures and Contraction Results

Lemma 1 (Bellman contraction under a stationary measure)**.** *Let P be a Markov operator with stationary distribution μ (i.e., $\mu = \mu P$), and define the Bellman operator $\mathcal{T} f := r + \gamma P f$. Then P is nonexpansive in $L^2(\mu)$:*

$$\|P(f-g)\|_{2,\mu} \;\le\; \|f-g\|_{2,\mu}.$$

Consequently, $\mathcal{T}$ is a γ-contraction:

$$\|\mathcal{T}(f-g)\|_{2,\mu} \;\le\; \gamma\,\|f-g\|_{2,\mu}.$$

Proof. Because μ is stationary for P, the operator P is nonexpansive in $L^2(\mu)$. Jensen's inequality implies

$$(Ph)^2 \le P(h^2) \qquad \text{pointwise.}$$

Integrating both sides with respect to μ and using $\mu = \mu P$ yields

$$\|Ph\|_{2,\mu}^2 = \int (Ph)^2\, d\mu \le \int P(h^2)\, d\mu = \int h^2\, d\mu = \|h\|_{2,\mu}^2.$$

Now apply this with $h = f - g$. Since the reward function r cancels,

$$\mathcal{T} f - \mathcal{T} g = \gamma\, P(f-g).$$

Therefore,

$$\|\mathcal{T} f - \mathcal{T} g\|_{2,\mu} = \gamma \|P(f-g)\|_{2,\mu} \le \gamma \|f-g\|_{2,\mu},$$

which proves the claim. □

Lemma 2 (Stationarity under coarsening)**.** *Let P be a Markov kernel on $\mathcal{S}$ and let ρ be stationary for P (i.e., $\rho P = \rho$). For any measurable coarsening map $g : \mathcal{S} \to \mathbb{R}$, define*

$$P_g f(s) := \mathbb{E}[\, f(S') \mid g(S) = g(s)\,],$$

where (S, S') has joint law $\rho(ds)\, P(s, ds')$. Then ρ is stationary for P_g.

Proof. Let $(S, S') \sim \rho(ds)\, P(s, ds')$, so $S \sim \rho$ and $S' \mid S \sim P(S, \cdot)$. Then

$$\begin{aligned}\int P_g f(s)\, \rho(ds) &= \mathbb{E}_{\rho,P}[P_g f(S)] \\ &= \mathbb{E}_{\rho,P}[\mathbb{E}_{\rho,P}[f(S') \mid g(S)]] \\ &= \mathbb{E}_{\rho,P}[f(S')].\end{aligned}$$

Since ρ is stationary for P, the marginal of S' is again ρ, hence $\mathbb{E}_{\rho,P}[f(S')] = \int f(s)\, \rho(ds)$. □

D. Proof of Theorem 1

Proof of Theorem 1. Define $P_{\pi,\hat{v}} := \Pi_{\hat{v}} P_\pi$ and $r_{\pi,\hat{v}} := \Pi_{\hat{v}} r_\pi$. Because v_0 and $\hat{v}_0$ satisfy the fixed-point equations

$$v_0 = \mathcal{T}_\pi v_0, \qquad \hat{v}_0 = \mathcal{T}_{\pi,\hat{v}} \hat{v}_0 = r_{\pi,\hat{v}} + \gamma\, P_{\pi,\hat{v}} \hat{v}_0,$$

we begin by writing

$$\hat{v}_0 - v_0 = \big(\mathcal{T}_{\pi,\hat{v}} \hat{v}_0 - \mathcal{T}_{\pi,\hat{v}} v_0\big) + \big(\mathcal{T}_{\pi,\hat{v}} v_0 - \mathcal{T}_\pi v_0\big).$$

Using linearity of the conditional expectation in $\mathcal{T}_{\pi,\hat{v}}$ on differences,

$$\mathcal{T}_{\pi,\hat{v}} \hat{v}_0 - \mathcal{T}_{\pi,\hat{v}} v_0 = \gamma\, P_{\pi,\hat{v}} (\hat{v}_0 - v_0),$$

so

$$\hat{v}_0 - v_0 = \gamma\, P_{\pi,\hat{v}} (\hat{v}_0 - v_0) + \big(\mathcal{T}_{\pi,\hat{v}} v_0 - v_0\big),$$

where we used $\mathcal{T}_\pi v_0 = v_0$ for the second term.

Taking $L^2(\mu_{\hat{v}})$ norms and applying the triangle inequality,

$$\begin{aligned}\|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} \le \gamma\, &\|P_{\pi,\hat{v}} (\hat{v}_0 - v_0)\|_{2,\mu_{\hat{v}}} \\ &+ \|\mathcal{T}_{\pi,\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}}.\end{aligned}$$

By Lemma 1,

$$\|P_{\pi,\hat{v}} h\|_{2,\mu_{\hat{v}}} \le \|h\|_{2,\mu_{\hat{v}}}.$$

Applying this with $h = \hat{v}_0 - v_0$ gives

$$\|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} \le \gamma\, \|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} + \|\mathcal{T}_{\pi,\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}}.$$

Note that $\mathcal{T}_{\pi,\hat{v}} v = \Pi_{\hat{v}} \mathcal{T}_\pi v$, where $\Pi_{\hat{v}}$ is the L^2 projection onto functions of $\hat{v}$. Thus $\mathcal{T}_{\pi,\hat{v}} v_0 = \Pi_{\hat{v}} v_0$, and therefore

$$\|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} \le \gamma\, \|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} + \|\Pi_{\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}}.$$

Rearranging,

$$\|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} \le \frac{1}{1-\gamma}\, \|\Pi_{\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}}. \qquad (\star)$$

By the assumed norm comparison,

$$\|\Pi_{\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}} \le \kappa_{\hat{v}} \|\Pi_{\hat{v}} v_0 - v_0\|_{2,\rho}.$$

Combining with $(\star)$ gives

$$\|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} \le \frac{\kappa_{\hat{v}}}{1-\gamma}\, \|\Pi_{\hat{v}} v_0 - v_0\|_{2,\rho}.$$

Finally, by the triangle inequality,

$$\|\hat{v} - v_0\|_{2,\mu_{\hat{v}}} \le \|\hat{v} - \hat{v}_0\|_{2,\mu_{\hat{v}}} + \|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}},$$

which yields the stated result. □

E. Proof of Theorem 2

Proof of Theorem 2. Fix v and treat the nuisance estimates as fixed functions. By definition,

$$\widehat{\mathcal{T}}_{0,\pi}(v)(s) = \mathbb{E}\Big[(\pi \hat{q}_v)(S) + \widehat{w}_\pi(A \mid S)\big\{R + \gamma\, v(S') - \hat{q}_v(S, A)\big\} \;\Big|\; S = s\Big].$$

Since $(\pi \hat{q}_v)(S)$ depends only on S,

$$\mathbb{E}\big[(\pi \hat{q}_v)(S) \mid S = s\big] = (\pi \hat{q}_v)(s).$$

For the second term, using the definition of b_0 and $q_v := r_0 + \gamma P v$,

$$\begin{aligned}&\mathbb{E}\Big[\widehat{w}_\pi(A \mid S)\big\{R + \gamma\, v(S') - \hat{q}_v(S, A)\big\} \;\Big|\; S = s\Big] \\ &\quad = b_0\Big\{\widehat{w}_\pi(\cdot \mid s)\, \mathbb{E}\big[R + \gamma\, v(S') - \hat{q}_v(s, \cdot) \mid S = s, A = \cdot\big]\Big\}(s) \\ &\quad = b_0\big\{\widehat{w}_\pi(q_v - \hat{q}_v)\big\}(s).\end{aligned}$$

Hence

$$\widehat{\mathcal{T}}_{0,\pi}(v)(s) = (\pi \hat{q}_v)(s) + b_0\big\{\widehat{w}_\pi(q_v - \hat{q}_v)\big\}(s).$$

Subtracting $\mathcal{T}_\pi(v)(s) = (\pi q_v)(s)$ yields

$$\begin{aligned}\widehat{\mathcal{T}}_{0,\pi}(v)(s) - \mathcal{T}_\pi(v)(s) &= (\pi \hat{q}_v - \pi q_v)(s) + b_0\big\{\widehat{w}_\pi(q_v - \hat{q}_v)\big\}(s) \\ &= (\pi(\hat{q}_v - q_v))(s) - b_0\big\{\widehat{w}_\pi(\hat{q}_v - q_v)\big\}(s).\end{aligned}$$

Using $(\pi f)(s) = b_0\{w_\pi f\}(s)$ with $f = \hat{q}_v - q_v$ gives

$$(\pi(\hat{q}_v - q_v))(s) = b_0\{w_\pi(\hat{q}_v - q_v)\}(s),$$

so

$$\widehat{\mathcal{T}}_{0,\pi}(v)(s) - \mathcal{T}_\pi(v)(s) = b_0\big\{(w_\pi - \widehat{w}_\pi)(\hat{q}_v - q_v)\big\}(s).$$

This proves the claim. □

F. Notation and Maximal Inequalities

Let P_0 denote the joint distribution of (S, A, R, S') induced by the behavior policy, and let P_n denote the empirical measure of the calibration sample $\mathcal{C}_n$.

We define empirical L^2 norms with respect to the state and next-state samples. For any state function f, let

$$\|f\|_{n,S} := \Big(\tfrac{1}{n}\sum_{i=1}^{n} f(S_i)^2\Big)^{1/2}, \quad \|f\|_{n,S'} := \Big(\tfrac{1}{n}\sum_{i=1}^{n} f(S_i')^2\Big)^{1/2},$$

where S_i and S_i' denote the observed states and next states, respectively, in the calibration sample $\mathcal{C}_n$.

Further define

$$\widehat{\mathcal{G}} := \Big\{(f_1 - f_2)\big(\widehat{\mathcal{T}}_\pi(f_2) - f_2\big) : \ f_1, f_2 \in \mathcal{F}_{B,\hat{v}}\Big\}.$$

By assumption, both $\hat{v}$ and $\widehat{\mathcal{T}}_\pi$ are fixed (non-random) operators conditional on the training data, which is independent of the calibration sample $\mathcal{C}_n$. Consequently, the classes $\mathcal{F}_{B,\hat{v}}$ and $\widehat{\mathcal{G}}$ are non-random conditional on the training dataset.

For any distribution Q and any uniformly bounded function class $\mathcal{F}$, let $N(\varepsilon, \mathcal{F}, L^2(Q))$ denote the ε-covering number of $\mathcal{F}$ under the $L^2(Q)$ norm (Van Der Vaart & Wellner, 1996). Define the uniform entropy integral of $\mathcal{F}$ by

$$\mathcal{J}(\delta, \mathcal{F}) := \int_0^\delta \sup_Q \sqrt{\log N(\epsilon, \mathcal{F}, L^2(Q))}\, d\epsilon,$$

where the supremum is taken over all discrete probability distributions Q.

Finally, for two quantities x and y, we write $x \lessapprox y$ to mean that x is bounded above by y up to a universal constant that depends only on global constants appearing in our conditions.

F.1. Local maximal inequality

Let $O_1, \ldots, O_n \in \mathcal{O}$ be independent random variables. For any function $f : \mathcal{O} \to \mathbb{R}$, define

$$\|f\|_{2,hob} := \sqrt{\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[f(O_i)^2]}. \tag{7}$$

For a star-shaped class of functions $\mathcal{F}$ and a radius $\delta \in (0, \infty)$, define the localized Rademacher complexity

$$\mathcal{R}_n(\mathcal{F}, \delta) := \mathbb{E}\left[\sup_{\substack{f \in \mathcal{F} \\ \|f\|_{2,hob} \le \delta}} \frac{1}{n}\sum_{i=1}^{n} \epsilon_i f(O_i)\right],$$

where ϵ_i are i.i.d. Rademacher random variables.

The following lemma provides a local maximal inequality and restates Lemma 11 of (Foster & Syrgkanis, 2023).

Lemma 3 (Local maximal inequality)**.** *Let $\mathcal{F}$ be a star-shaped class of functions satisfying $\sup_{f\in\mathcal{F}} \|f\|_\infty \le M$. Let $\delta > 0$ satisfy the critical radius condition $\mathcal{R}_n(\mathcal{F}, \delta) \le \delta^2$. Suppose further that $n^{-1/2}\sqrt{\log\log(1/\delta)} = o(\delta)$. Then there exists a universal constant $C > 0$ such that, for all $u \ge 1$, with probability at least $1 - e^{-u^2}$, every $f \in \mathcal{F}$ satisfies*

$$\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big) \le C\Big(\delta^2 + \delta\|f\|_{2,hob} + \frac{u\|f\|_{2,hob}}{\sqrt{n}} + \frac{Mu^2}{n}\Big).$$

The following lemma bounds the localized Rademacher complexity in terms of the uniform entropy integral and is a direct consequence of Theorem 2.1 of Van Der Vaart & Wellner (2011).

Lemma 4. *Let $\mathcal{F}$ be a star-shaped class of functions such that $\sup_{f\in\mathcal{F}} \|f\|_\infty \le M$. Then, for every $\delta > 0$,*

$$\mathcal{R}_n(\mathcal{F}, \delta) \lesssim \frac{1}{\sqrt{n}}\mathcal{J}(\delta, \mathcal{F})\left(1 + \frac{\mathcal{J}(\delta, \mathcal{F})}{\delta\sqrt{n}}\right),$$

where the implicit constant depends only on M.

Proof. This bound follows directly from the argument in the proof of Theorem 2.1 of Van Der Vaart & Wellner (2011); see in particular the step where the local Rademacher complexity is controlled by the uniform entropy integral for star-shaped classes. □

G. Proof of Theorem 3

G.1. Technical lemmas

Lemma 5. *Under our conditions,*

$$\mathcal{J}\left(\delta, \widehat{\mathcal{G}}\right) \lesssim \delta\sqrt{B\log(1/\delta)},$$

where the implicit constant is independent of B.

Proof. By assumption, $\|\widehat{P}f - \widehat{P}g\|_{n,A,S} \le L\|f - g\|_{n,S'}$ almost surely. For each s, by Jensen's inequality,

$$|\pi h(s)|^2 = \Big|\sum_a \pi(a \mid s)h(a, s)\Big|^2 \le \sum_a \pi(a \mid s)|h(a, s)|^2,$$

so $\|\pi h\|_{n,S} \le \|h\|_{n,A,S}$. Hence, we also have

$$\|\pi\widehat{P}f - \pi\widehat{P}g\|_{n,S} \le \|\widehat{P}f - \widehat{P}g\|_{n,A,S} \le L\|f - g\|_{n,S'}.$$

Hence, by boundedness of all nuisances,

$$\|\widehat{\mathcal{T}}_\pi(f) - \widehat{\mathcal{T}}_\pi(g)\|_{L^2(P_n)} \lesssim \|f - g\|_{n,S} + \|f - g\|_{n,S'}.$$

Take $f, g \in \widehat{\mathcal{G}}$ with $f = (f_1 - f_2)\big(\widehat{\mathcal{T}}_\pi(f_2) - f_2\big)$ and $g = (g_1 - g_2)\big(\widehat{\mathcal{T}}_\pi(g_2) - g_2\big)$ for $f_1, f_2, g_1, g_2 \in \mathcal{F}_{B,\hat{v}}$. By

Lipschitz continuity of multiplication and boundedness of nuisances, we have

$$\|f-g\|_{L^2(P_n)} \lesssim \|f_1-g_1\|_{n,S}+\|f_2-g_2\|_{n,S}+\|f_2-g_2\|_{n,S'}.$$

Taking the supremum over all discrete distributions Q, we obtain

$$\|f-g\|_{L^2(P_n)} \lesssim \sup_Q \|f_1-g_1\|_{L^2(Q)}+\sup_Q \|f_2-g_2\|_{L^2(Q)}.$$

Hence, by preservation of entropy integrals in Van Der Vaart & Wellner (1996),

$$\log N(\varepsilon, \widehat{\mathcal{G}}, L^2(P_n)) \lesssim \sup_Q \log N(\varepsilon, \mathcal{F}_{B,\hat{v}}, L^2(Q)).$$

Taking the supremum over discrete distributions Q on both sides yields the uniform covering number bound

$$\sup_Q \log N(\varepsilon, \widehat{\mathcal{G}}, L^2(Q)) \lesssim \sup_Q \log N(\varepsilon, \mathcal{F}_{B,\hat{v}}, L^2(Q)).$$

The class $\mathcal{F}_{B,\hat{v}}$ satisfies

$$\sup_Q \log N\big(\varepsilon, \mathcal{F}_{B,\hat{v}}, L^2(Q)\big) \;\le\; \sup_Q \log N\big(\varepsilon, \widetilde{\mathcal{F}}_B, L^2(Q\circ\hat{v}^{-1})\big),$$

where $Q \circ \hat{v}^{-1}$ denotes the pushforward of Q under $\hat{v}$. Hence,

$$\sup_Q \log N\big(\varepsilon, \mathcal{F}_{B,\hat{v}}, L^2(Q)\big) \;\le\; \sup_Q \log N\big(\varepsilon, \widetilde{\mathcal{F}}_B, L^2(Q)\big),$$

where, by a slight abuse of notation, the supremum on the right-hand side is taken over all discrete probability distributions Q on $\mathbb{R}$. The class $\widetilde{\mathcal{F}}_B$ consists of all piecewise-constant functions on $\mathbb{R}$ taking at most B values. Therefore, $\widetilde{\mathcal{F}}_B$ has VC–subgraph dimension $O(B)$, and Van Der Vaart & Wellner (1996) implies

$$\begin{aligned}\sup_Q \log N(\varepsilon, \mathcal{F}_{B,\hat{v}}, L^2(Q)) &\lesssim \sup_Q \log N\big(\varepsilon, \widetilde{\mathcal{F}}_B, L^2(Q)\big)\\ &\lesssim B\log(1/\varepsilon).\end{aligned}$$

Consequently, $\mathcal{J}(\delta, \mathcal{F}_{B,\hat{v}}) \lesssim \delta\sqrt{B\log(1/\delta)}$. □

G.2. Proof of Theorem 3

Proof of Theorem 3. Denote $O_i := (S_i, A_i, R_i, S_i')$. The first-order optimality conditions for $\theta_n^{(K)}$ imply that, for each bin $b \in [B]$,

$$\frac{1}{n}\sum_{i=1}^n \mathbf{1}_{I_b}(\hat{v}(S_i))\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})(O_i) - \hat{v}^{(K)}(S_i)\} = 0.$$

Hence, for any $f \in \mathcal{F}_B$, which is a linear combination of these indicators,

$$\frac{1}{n}\sum_{i=1}^n f(\hat{v}(S_i))\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})(O_i) - \hat{v}^{(K)}(S_i)\} = 0. \quad (8)$$

Moreover, for any function $g : \mathbb{R} \to \mathbb{R}$, we may write $g \circ \hat{v}^{(K)} = f \circ \hat{v}$ for some f, and therefore, for all such g,

$$\frac{1}{n}\sum_{i=1}^n g(\hat{v}^{(K)}(S_i))\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})(O_i) - \hat{v}^{(K)}(S_i)\} = 0.$$

Noting that $\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}$ is of the form $g \circ \hat{v}^{(K)}$ for some function g, we have

$$\begin{aligned}0 = \frac{1}{n}\sum_{i=1}^n &\Big\{\Gamma_0(\hat{v}^{(K)})(S_i) - \hat{v}^{(K)}(S_i)\Big\}\\ &\times \Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})(O_i) - \hat{v}^{(K)}(S_i)\Big\}.\end{aligned}$$

Adding and subtracting P_0 yields

$$\begin{aligned}&P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\\ &= (P_0 - P_n)\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) - \hat{v}^{(K)}\Big\}\\ &\quad + P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})\Big\}.\end{aligned} \quad (9)$$

We rewrite the left-hand side of (9). Adding and subtracting and applying the law of total expectation, the calibration error decomposes as

$$\begin{aligned}&\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|^2\\ &= P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\\ &\quad + P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\Gamma_0(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_\pi(\hat{v}^{(K)})\Big\}.\end{aligned} \quad (10)$$

The first term on the right-hand side can be decomposed as in (9). By the law of total expectation, the proof of Theorem 2, and the Cauchy–Schwarz inequality, the second term satisfies

$$\begin{aligned}&\Big|P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \Gamma_0(\hat{v}^{(K)})\Big\}\Big|\\ &= \Big|P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \mathcal{T}_\pi(\hat{v}^{(K)})\Big\}\Big|\\ &\le \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|\;\big\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\big\|_{2,\rho}.\end{aligned}$$

Hence, (9) and (10) together imply that

$$\begin{aligned}&\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|^2\\ &\le\; (P_0 - P_n)\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) - \hat{v}^{(K)}\Big\}\\ &\quad + \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|\;\big\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\big\|\\ &\quad + \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|\;\big\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\big\|_{2,\rho}.\end{aligned} \quad (11)$$

where $\widehat{\mathcal{T}}_{0,\pi}(v)(s) := \mathbb{E}[\widehat{\mathcal{T}}_\pi(v)(S) \mid S = s]$. Here, the second term on the right-hand side follows from (9), noting that

$$\begin{aligned}
&\left|P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})\Big\}\right| \\
&= \left|P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\,\mathbb{E}\left[\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) \mid S\right]\right| \\
&\le \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \left\|\mathbb{E}\left[\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) \mid S\right]\right\| \\
&= \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \left\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\right\|_{2,\rho}.
\end{aligned}$$

We now turn to bounding the empirical process term on the right-hand side of (11). Observe that

$$\big(\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\big)\big(\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) - \hat{v}^{(K)}\big)$$

lies in a uniformly bounded subset of the class

$$\widehat{\mathcal{G}} := \big\{(f_1 - f_2)\big(\widehat{\mathcal{T}}_\pi(f_2) - f_2\big) : f_1, f_2 \in \mathcal{F}_{B,\hat{v}}\big\},$$

By Lemma 5, it holds that $\mathcal{J}\left(\delta, \widehat{\mathcal{G}}\right) \lesssim \delta\sqrt{B\log(1/\delta)}$. By assumption, $\widehat{\mathcal{G}}$ is a fixed, nonrandom function class conditional on the training data, which is independent of the calibration sample $\mathcal{C}_n$. Hence, applying Lemma 4 conditional on the training data, we obtain

$$\begin{aligned}
\mathcal{R}_n(\widehat{\mathcal{G}}, \delta) &\lesssim \frac{1}{\sqrt{n}}\,\mathcal{J}(\delta, \widehat{\mathcal{G}})\left(1 + \frac{\mathcal{J}(\delta, \widehat{\mathcal{G}})}{\delta\sqrt{n}}\right) \\
&\lesssim \frac{1}{\sqrt{n}}\,\delta\sqrt{B\log(1/\delta)}\left(1 + \frac{\delta\sqrt{B\log(1/\delta)}}{\delta\sqrt{n}}\right) \\
&\lesssim \frac{\delta\sqrt{B\log(1/\delta)}}{\sqrt{n}} + \frac{B\log(1/\delta)}{n}.
\end{aligned}$$

The critical radius

$$\delta_n := \sqrt{\frac{B}{n}\log\left(\frac{n}{B}\right)},$$

satisfies

$$\delta_n = \inf\{\delta > 0 : \mathcal{R}_n(\widehat{\mathcal{G}}, \delta) \lesssim \delta^2\}.$$

Applying Lemma 3 conditional on the training data with $\mathcal{F} := \widehat{\mathcal{G}}$, we conclude that the following holds with probability at least $1 - e^{-u^2}$ for every $f \in \widehat{\mathcal{G}}$:

$$\frac{1}{n}\sum_{i=1}^n \big(f(O_i) - \mathbb{E}[f(O_i)]\big) \lesssim \delta_n^2 + \delta_n\|f\|_{2,hob} + \frac{u\|f\|_{2,hob}}{\sqrt{n}} + \frac{u^2}{n}.$$

Choosing $u = \sqrt{\log(1/\eta)}$ gives $1 - e^{-u^2} = 1 - \eta$. Hence, with probability at least $1 - \eta$,

$$\frac{1}{n}\sum_{i=1}^n \big(f(O_i) - \mathbb{E}[f(O_i)]\big) \lesssim \delta_n^2 + \delta_n\|f\|_{2,hob} + \frac{\sqrt{\log(1/\eta)}\,\|f\|_{2,hob}}{\sqrt{n}} + \frac{\log(1/\eta)}{n}.$$

By boundedness of our nuisances,

$$\left\|\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\}\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) - \hat{v}^{(K)}\}\right\| \lesssim \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|_{2,\rho}.$$

Hence, with probability at least $1 - \eta$,

$$\begin{aligned}
(P_n - P_0)\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) - \hat{v}^{(K)}\Big\} &\lesssim \delta_n^2 + \delta_n\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \\
&\quad + \frac{\sqrt{\log(1/\eta)}\,\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|}{\sqrt{n}} + \frac{\log(1/\eta)}{n},
\end{aligned}$$

where the implicit constants do not depend on B.

Combining the above with (11), we find with probability at least $1 - \eta$,

$$\begin{aligned}
&\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|^2 \\
&\lesssim \delta_n^2 + \delta_n\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \\
&\quad + \frac{\sqrt{\log(1/\eta)}}{\sqrt{n}}\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| + \frac{\log(1/\eta)}{n} \\
&\quad + \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|\,\left\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\right\| \\
&\quad + \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|\,\left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\right\|_{2,\rho}.
\end{aligned} \tag{12}$$

The inequality in (12) implies that, with probability at least $1 - \eta$, the calibration error satisfies

$$\begin{aligned}
\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| &\lesssim \delta_n + \sqrt{\frac{\log(1/\eta)}{n}} \\
&\quad + \left\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\right\| \\
&\quad + \left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\right\|_{2,\rho}.
\end{aligned}$$

Recall that $\delta_n := \sqrt{\frac{B}{n}\log\left(\frac{n}{B}\right)}$. Then, with probability at least $1 - \eta$,

$$\begin{aligned}
\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| &\lesssim \sqrt{\frac{B}{n}\log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(1/\eta)}{n}} \\
&\quad + \left\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\right\| \\
&\quad + \left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\right\|_{2,\rho}.
\end{aligned}$$

□

H. Proof of Theorem 4

H.1. Additional notation

Define the bin–index map $\hat{b}(s)$ by setting $\hat{b}(s) = b$ whenever $\hat{v}(s) \in I_b$. Let $\Pi_{\hat{v},B}$ denote the binning projection

$$(\Pi_{\hat{v},B} g)(s) := E\Big[g(S) \,\Big|\, \hat{v}(S) \in I_{\hat{b}(s)}\Big].$$

Denote

$$\mathcal{T}_{\pi,\hat{v},B}(g)(s) := (\Pi_{\hat{v},B}\mathcal{T}_\pi(g)(s)$$

Define $\hat{v}_{0,B}$ denote the fixed point satisfying

$$\mathcal{T}_{\pi,\hat{v},B}(\hat{v}_{0,B}) = \hat{v}_{0,B}.$$

Such a fixed point exists by Banach's fixed point theorem because $\mathcal{T}_{\pi,\hat{v},B}(g) \in \mathcal{F}_{B,\hat{v}}$ whenever $g \in \mathcal{F}_{B,\hat{v}}$, and the operator is a γ-contraction in the sup-norm.

Recall the projected transition operator

$$(P_{\pi,\hat{v},B}h)(s) = \mathbb{E}_\pi\left[h(S') \,\Big|\, \hat{v}(S) \in I_{\hat{b}(s)}\right].$$

It holds that

$$P_{\pi,\hat{v},B} \;:=\; \Pi_{\hat{v},B}P_\pi,$$

and

$$\Pi_{\hat{v},B}\mathcal{T}_\pi(f) - \Pi_{\hat{v},B}\mathcal{T}_\pi(g) = \gamma\, P_{\pi,\hat{v},B}(f-g).$$

H.2. Supporting lemmas

We begin by bounding the error between the projected fixed point $\hat{v}_{0,B}$ and the true value function v_0.

Lemma 6 (Approximation error for the projected fixed point)**.** *Assume that $\mu_{\hat{v},B}$ is stationary for $P_{\pi,\hat{v},B}$. Then*

$$\|\hat{v}_{0,B} - v_0\|_{L^2(\mu_{\hat{v},B})} \;\le\; \frac{1}{1-\gamma}\,\|(I-\Pi_{\hat{v},B})v_0\|_{L^2(\mu_{\hat{v},B})}.$$

Proof. Because v_0 and $\hat{v}_{0,B}$ satisfy $v_0 = \mathcal{T}_\pi v_0$ and $\hat{v}_{0,B} = \Pi_{\hat{v},B}\mathcal{T}_\pi\hat{v}_{0,B}$, we have

$$\hat{v}_{0,B} - v_0 = \Pi_{\hat{v},B}\big(\mathcal{T}_\pi\hat{v}_{0,B} - \mathcal{T}_\pi v_0\big) + (I-\Pi_{\hat{v},B})\mathcal{T}_\pi v_0.$$

Since $\mathcal{T}_\pi(f) - \mathcal{T}_\pi(g) = \gamma P_\pi(f-g)$ and $\Pi_{\hat{v},B}$ is affine but linear on differences, the first term equals

$$\begin{aligned}\Pi_{\hat{v},B}\big(\mathcal{T}_\pi\hat{v}_{0,B} - \mathcal{T}_\pi v_0\big) &= \gamma\,\Pi_{\hat{v},B}P_\pi(\hat{v}_{0,B} - v_0)\\ &= \gamma\, P_{\pi,\hat{v},B}(\hat{v}_{0,B} - v_0).\end{aligned}$$

Using $\mathcal{T}_\pi v_0 = v_0$, we therefore obtain the error identity

$$\hat{v}_{0,B} - v_0 = \gamma\, P_{\pi,\hat{v},B}(\hat{v}_{0,B} - v_0) + (I-\Pi_{\hat{v},B})v_0.$$

Taking $L^2(\mu_{\hat{v},B})$ norms and applying the triangle inequality yields

$$\begin{aligned}\|\hat{v}_{0,B} - v_0\|_{2,\mu_{\hat{v},B}} \le \gamma\,\|P_{\pi,\hat{v},B}(\hat{v}_{0,B} - v_0)\|_{2,\mu_{\hat{v},B}}\\ +\; \|(I-\Pi_{\hat{v},B})v_0\|_{2,\mu_{\hat{v},B}}.\end{aligned}$$

Because $P_{\pi,\hat{v},B}$ is a Markov operator and $\mu_{\hat{v},B}$ is stationary for $P_{\pi,\hat{v},B}$, it is nonexpansive in $L^2(\mu_{\hat{v},B})$. By Jensen's inequality, $(P_{\pi,\hat{v},B}h)^2 \le P_{\pi,\hat{v},B}(h^2)$ pointwise, hence

$$\begin{aligned}\|P_{\pi,\hat{v},B}h\|^2_{L^2(\mu_{\hat{v},B})} &= \int (P_{\pi,\hat{v},B}h)^2\, d\mu_{\hat{v},B}\\ &\le \int P_{\pi,\hat{v},B}(h^2)\, d\mu_{\hat{v},B}\\ &= \int h^2\, d\mu_{\hat{v},B}\\ &= \|h\|^2_{L^2(\mu_{\hat{v},B})}.\end{aligned}$$

where the equality uses stationarity of $\mu_{\hat{v},B}$. Applying this with $h = \hat{v}_{0,B} - v_0$ gives $\|P_{\pi,\hat{v},B}(\hat{v}_{0,B} - v_0)\|_{L^2(\mu_{\hat{v},B})} \le \|\hat{v}_{0,B} - v_0\|_{L^2(\mu_{\hat{v},B})}$. Therefore,

$$\|\hat{v}_{0,B} - v_0\|_{L^2(\mu_{\hat{v},B})} \le \gamma\,\|\hat{v}_{0,B} - v_0\|_{L^2(\mu_{\hat{v},B})} + \|(I-\Pi_{\hat{v},B})v_0\|_{L^2(\mu_{\hat{v},B})}.$$

Rearranging completes the proof:

$$\|\hat{v}_{0,B} - v_0\|_{L^2(\mu_{\hat{v},B})} \le \frac{1}{1-\gamma}\,\|(I-\Pi_{\hat{v},B})v_0\|_{L^2(\mu_{\hat{v},B})}.$$

□

Lemma 7 (Inexact iterations of $\mathcal{T}_{\pi,\hat{v},B}$)**.** *Assume $\mu_{\hat{v},B}$ is stationary for $P_{\pi,\hat{v},B}$, so $\mu_{\hat{v},B} = \mu_{\hat{v},B}P_{\pi,\hat{v},B}$. Let $\{\eta_k\}_{k\ge1}$ be any sequence such that*

$$\|\hat{v}^{(k)} - \mathcal{T}_{\pi,\hat{v},B}(\hat{v}^{(k-1)})\|_{2,\mu_{\hat{v},B}} \;\le\; \eta_k \qquad \textit{for all } k.$$

Then, for any $K \ge 1$,

$$\|\hat{v}^{(K)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \;\le\; \gamma^K\,\|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} + \sum_{j=1}^{K}\gamma^{K-j}\,\eta_j.$$

Proof. Because $\mathcal{F}_B$ consists of functions that are constant on each bin, conditioning on $\hat{b}(S)$ yields a function that is also constant on bins. Thus $\mathcal{T}_{\pi,\hat{v},B}$ maps $\mathcal{F}_B$ into itself.

Define the per-iteration error function

$$e_k := \hat{v}^{(k)} - \mathcal{T}_{\pi,\hat{v},B}(\hat{v}^{(k-1)}), \qquad \|e_k\|_{2,\mu_{\hat{v},B}} \le \eta_k.$$

Since $\hat{v}^{(k)}, \hat{v}_{0,B} \in \mathcal{F}_B$, there exist $\hat{\theta}^{(k)}$ and $\theta^\star$ such that $\hat{v}^{(k)} = \hat{\theta}^{(k)} \circ \hat{v}$ and $\hat{v}_{0,B} = \theta^\star \circ \hat{v}$. By Lemma 1, $\mathcal{T}_{\pi,\hat{v},B}$ is a γ-contraction under $\|\cdot\|_{2,\mu_{\hat{v},B}}$. Applying the standard inexact-iteration bound (e.g., Lemma 4 of (van der Laan et al., 2025b)) gives

$$\|(\hat{\theta}^{(K)} - \theta^\star)\circ\hat{v}\|_{2,\mu_{\hat{v},B}} \;\le\; \gamma^K\|(\hat{\theta}^{(0)} - \theta^\star)\circ\hat{v}\|_{2,\mu_{\hat{v},B}} + \sum_{j=1}^{K}\gamma^{K-j}\,\|e_j\|_{2,\mu_{\hat{v},B}}.$$

Substituting $\hat{v}^{(k)} = \hat{\theta}^{(k)} \circ \hat{v}$ and $\hat{v}_{0,B} = \theta^\star \circ \hat{v}$ yields

$$\|\hat{v}^{(K)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \;\le\; \gamma^K\,\|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} + \sum_{j=1}^{K}\gamma^{K-j}\,\eta_j.$$

□

Lemma 8 (Error bound for inexact updates)**.** *There exists a $C < \infty$, such that, with probability at least $1-\delta$, it holds that, uniformly over $k \in [K]$,*

$$\|\hat{v}^{(k)} - \mathcal{T}_{\pi,\hat{v},B}(v^{(k-1)})\| \leq \sqrt{\frac{B}{n}\log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(K/\delta)}{n}} + C\left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}})\right\|_{2,\rho}.$$

Proof. Equation 8 in the proof of Theorem 3 shows that, for any transformation $f \in \mathcal{F}_B$ and any k,

$$\frac{1}{n}\sum_{i=1}^{n} f(\hat{v}(S_i))\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(k-1)})(O_i) - \hat{v}^{(k)}(S_i)\} = 0.$$

Applying the above with $f \circ \hat{v} = \hat{v}_B^{\star(k)} - \hat{v}^{(k)}$, where

$$\hat{v}_B^{\star(k)} := \mathcal{T}_{\pi,\hat{v},B}\left(\theta^{(k-1)}\right),$$

we obtain

$$\frac{1}{n}\sum_{i=1}^{n}(\hat{v}_B^{\star(k)} - \hat{v}^{(K)})(S_i)\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(K-1)}\right)(O_i) - \hat{v}^{(K)}(S_i)\right\} = 0.$$

Adding and subtracting P_0, we find

$$P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}^{(k)}\right\}\right] = (P_0 - P_n)\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}^{(k)}\right\}\right]. \tag{13}$$

First, we study the left–hand side of (13). We have

$$P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}^{(k)}\right\}\right] = \|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|^2 + P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}_B^{\star(k)}\right\}\right].$$

By the law of total expectation,

$$P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}_B^{\star(k)}\right\}\right] = P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \mathcal{T}_{\pi,\hat{v},B}\left(\theta^{(k-1)}\right)\right\}\right] = P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \mathcal{T}_\pi\left(\hat{v}^{(k-1)}\right)\right\}\right].$$

By Theorem 2 and arguing as in the proof of Theorem 3, it follows that

$$\left|P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}_B^{\star(k)}\right\}\right]\right| \leq \|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|\,\left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}})\right\|_{2,\rho}.$$

Putting it all together,

$$\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|^2 \leq P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}^{(k)}\right\}\right] + \|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|\,\left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}})\right\|_{2,\rho}.$$

Hence, by (13),

$$\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|^2 \leq (P_0 - P_n)\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}^{(k)}\right\}\right] + \|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|\,\left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}})\right\|_{2,\rho}. \tag{14}$$

Next, we obtain a high–probability bound for the first term right–hand side of (14). Applying Lemmas 3, 4, and 5, and arguing as in the proof of Theorem 3, we obtain that, with probability at least $1-\delta$,

$$\left|(P_0 - P_n)\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(k-1)}) - \hat{v}^{(k)}\}\right]\right| \lesssim \delta_n^2 + \delta_n\,\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\| + \frac{\sqrt{\log(1/\delta)}\,\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|}{\sqrt{n}} + \frac{\log(1/\delta)}{n},$$

where $\delta_n := \sqrt{(B/n)\,\log(B/\delta)}$, and the implicit constants do not depend on B.

Plugging this high probability bound into (14), we obtain that, with probability at least $1-\delta$,

$$\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|^2 \lesssim \delta_n^2 + \delta_n\,\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\| + \frac{\sqrt{\log(1/\delta)}\,\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|}{\sqrt{n}} + \frac{\log(1/\delta)}{n} + \|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|\,\left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}})\right\|_{2,\rho}.$$

Recalling that $\delta_n := \sqrt{\frac{B}{n}\log\left(\frac{n}{B}\right)}$, the inequality above implies that, with probability at least $1-\delta$,

$$\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\| \lesssim \sqrt{\frac{B}{n}\log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(1/\delta)}{n}} + \left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}})\right\|_{2,\rho}.$$

Recalling that $\eta_k = \hat{v}_B^{\star(k)}$, the result follows by a union bound over $k \in [K]$. □

H.3. Proof of Theorem 4

Proof of Theorem 4. For $C \in (0, \infty)$ large enough, define

$$\eta_k = C\left(\sqrt{\frac{B}{n}\log\Big(\frac{n}{B}\Big)} + \sqrt{\frac{\log(K/\delta)}{n}}\right) + \left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}})\right\|_{2,\rho}.$$

By Lemma 8 and the norm bound in C4, for C large enough, with probability at least $1-\delta$ and uniformly over $k \in [K]$,

$$\|\hat{v}^{(k)} - \mathcal{T}_{\pi,\hat{v},B}(\hat{v}^{(k-1)})\|_{2,\mu_{\hat{v},B}} \le \hat{\kappa}\,\|\hat{v}^{(k)} - \mathcal{T}_{\pi,\hat{v},B}(\hat{v}^{(k-1)})\| \le \hat{\kappa}\,\eta_k.$$

Define $\tilde{\eta}_k := \hat{\kappa}\,\eta_k$. Then the above shows that the assumption of Lemma 7 holds with $\tilde{\eta}_k$ in place of η_k.

Applying the deterministic inequality of Lemma 7 with $\tilde{\eta}_k$ yields, with probability at least $1-\delta$,

$$\begin{aligned}
&\|\hat{v}^{(K)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\le \gamma^K\,\|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} + \sum_{j=1}^{K}\gamma^{K-j}\,\tilde{\eta}_j \\
&\le \gamma^K\,\|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\quad + \frac{C\hat{\kappa}}{1-\gamma}\left(\sqrt{\frac{B}{n}\log\Big(\frac{n}{B}\Big)} + \sqrt{\frac{\log(K/\delta)}{n}}\right) \\
&\quad + \frac{\hat{\kappa}}{1-\gamma}\max_{j\in[K]}\left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(j-1)}} - q_{\hat{v}^{(j-1)}})\right\|_{2,\rho}.
\end{aligned}$$

By Lemma 6,

$$\|\hat{v}_{0,B} - v_0\|_{2,\mu_{\hat{v},B}} \le \frac{1}{1-\gamma}\,\|(I - \Pi_{\hat{v},B})v_0\|_{2,\mu_{\hat{v},B}}.$$

Combining this with our high–probability bound on $\|\hat{v}^{(K)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}}$, we find that, with probability at least $1-\delta$,

$$\begin{aligned}
&\|\hat{v}^{(K)} - v_0\|_{2,\mu_{\hat{v},B}} \\
&\le \frac{1}{1-\gamma}\|\Pi_{\hat{v},B}v_0 - v_0\|_{2,\mu_{\hat{v},B}} \\
&\quad + \gamma^K\,\|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\quad + \frac{C\hat{\kappa}}{1-\gamma}\left(\sqrt{\frac{B}{n}\log\Big(\frac{n}{B}\Big)} + \sqrt{\frac{\log(K/\delta)}{n}}\right) \\
&\quad + \frac{\hat{\kappa}}{1-\gamma}\max_{j\in[K]}\left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(j-1)}} - q_{\hat{v}^{(j-1)}})\right\|_{2,\rho}.
\end{aligned}$$

Since $\Pi_{\hat{v},B}$ is a projection under $\|\cdot\|_{2,\rho}$, we have

$$\begin{aligned}
\|\hat{v}_{0,B} - v_0\|_{2,\mu_{\hat{v},B}} &\le \frac{\hat{\kappa}}{1-\gamma}\,\|(I - \Pi_{\hat{v},B})v_0\| \\
&\le \frac{\hat{\kappa}}{1-\gamma}\min_{\theta\in\mathcal{F}_B}\|\theta\circ\hat{v} - v_0\|,
\end{aligned}$$

where we used the second norm bound in C4. □

H.4. Bound on Successive Iteration Errors

Lemma 9 (Bound on Successive Iteration Errors)**.** *Under the conditions of Theorem 4,*

$$\begin{aligned}
&\|\hat{v}^{(K)} - \hat{v}^{(K-1)}\|_{2,\mu_{\hat{v},B}} \\
&\lesssim \gamma^K\,\|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\quad + \frac{C}{1-\gamma}\left(\sqrt{\frac{B}{n}\log\Big(\frac{n}{B}\Big)} + \sqrt{\frac{\log(K/\delta)}{n}}\right) \\
&\quad + \frac{\kappa_{\hat{v},B}}{1-\gamma}\max_{j\in[K]}\left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(j-1)}} - q_{\hat{v}^{(j-1)}})\right\|_{2,\rho}.
\end{aligned}$$

Proof. The proof of Theorem 4 established that

$$\begin{aligned}
&\|\hat{v}^{(K)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\le \gamma^K\,\|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\quad + \frac{C\kappa_{\hat{v},B}}{1-\gamma}\left(\sqrt{\frac{B}{n}\log\Big(\frac{n}{B}\Big)} + \sqrt{\frac{\log(K/\delta)}{n}}\right) \\
&\quad + \frac{\kappa_{\hat{v},B}}{1-\gamma}\max_{j\in[K]}\left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(j-1)}} - q_{\hat{v}^{(j-1)}})\right\|_{2,\rho}.
\end{aligned}$$

By the triangle inequality,

$$\begin{aligned}
\|\hat{v}^{(K)} - \hat{v}^{(K-1)}\|_{2,\mu_{\hat{v},B}} &\le \|\hat{v}^{(K-1)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\quad + \|\hat{v}^{(K)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}}.
\end{aligned}$$

The result follows by applying the above bound with $K-1$ and K. □

I. Proofs of Theorems 5 and 6

Proof of Theorem 5. The first-order optimality conditions for isotonic regression (equivalently, its interpretation as a histogram estimator) imply that for any function $f : \mathbb{R} \to \mathbb{R}$ that is a linear combination of the indicator functions defining the isotonic partition,

$$\frac{1}{n}\sum_{i=1}^{n} f\big(\hat{v}^{(K)}(S_i)\big)\Big\{\widehat{\mathcal{T}}_\pi\big(\hat{v}^{(K-1)}\big)(O_i) - \hat{v}^{(K)}(S_i)\Big\} = 0.$$

See, for example, the proof of Lemma C.1 in Van Der Laan et al. (2023) and van der Laan et al. (2024). Choosing f appropriately yields

$$0 = \frac{1}{n}\sum_{i=1}^{n}\Big\{\Gamma_0(\hat{v}^{(K)})(S_i) - \hat{v}^{(K)}(S_i)\Big\} \times \Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})(O_i) - \hat{v}^{(K)}(S_i)\Big\},$$

which is the same basic equality as (9) in the proof of Theorem 3. The remainder of the argument proceeds along the same lines with minor modifications.

Specifically, let $\mathcal{F}_{TV}$ denote the union of $\mathcal{F}_{iso}$, which is uniformly bounded by $2M$ under Condition C1, with all functions of bounded total variation bounded by the constant C in Condition C5. By Van Der Vaart & Wellner (1996), this class satisfies the uniform entropy integral bound $\mathcal{J}(\delta, \mathcal{F}_{TV}) \lesssim \sqrt{\delta}$.

Under Condition C5 and by Lemma 6 of van der Laan et al. (2024), $\Gamma_0(\hat{v}^{(K)})$ has finite total variation bounded and lies in $\mathcal{F}_{TV,\hat{v}} := \{f \circ \hat{v} : f \in \mathcal{F}_{TV}\}$. Thus, it holds that

$$\big(\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\big)\big(\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) - \hat{v}^{(K)}\big)$$

lies in a uniformly bounded subset of the class

$$\widehat{\mathcal{G}} := \Big\{(f_1 - f_2)\big(\widehat{\mathcal{T}}_\pi(f_2) - f_2\big) : \ f_1, f_2 \in \mathcal{F}_{TV,\hat{v}}\Big\}.$$

Arguing as in the proof of Theorem 3, we have

$$\mathcal{J}(\delta, \widehat{\mathcal{G}}) \lesssim \mathcal{J}(\delta, \mathcal{F}_{TV}) \lesssim \sqrt{\delta}.$$

The proof now follows directly from Theorem 3 with this new choice of $\widehat{\mathcal{G}}$ and its associated critical radius $\delta_n = n^{-1/3}$ for monotone functions. □

Proof of Theorem 6. The result follows directly from a union bound and the proofs of Theorems 3 and 4, with $\mathcal{F}_B$ replaced by $\mathcal{F}_{B_n}$, the class of all piecewise-constant functions with at most B_n constant segments. In particular, the entropy bound in Lemma 5 continues to apply to this class. Notably, the proofs of these theorems allow for data-adaptive partitions and require only a deterministic upper bound on the number of constant segments. □